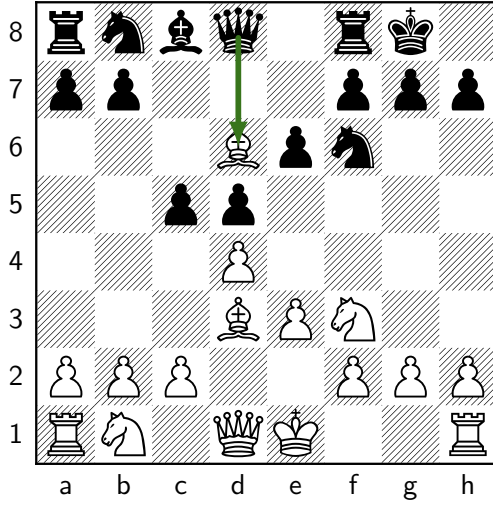


# Melanie-ANALOGY — London System

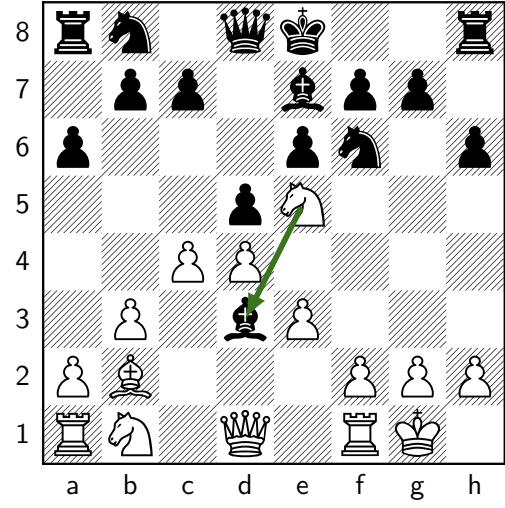
Melanie (Deep\_Vision)

## Abstract

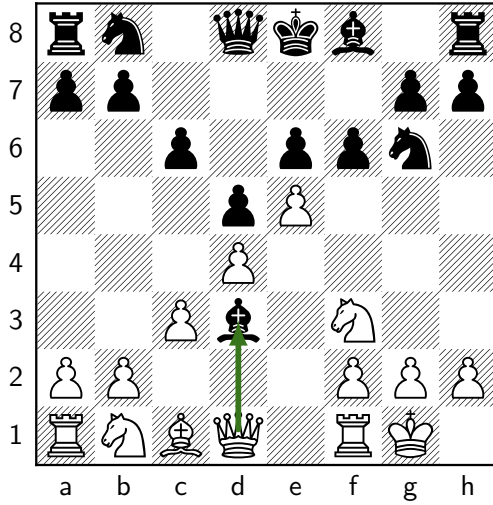
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



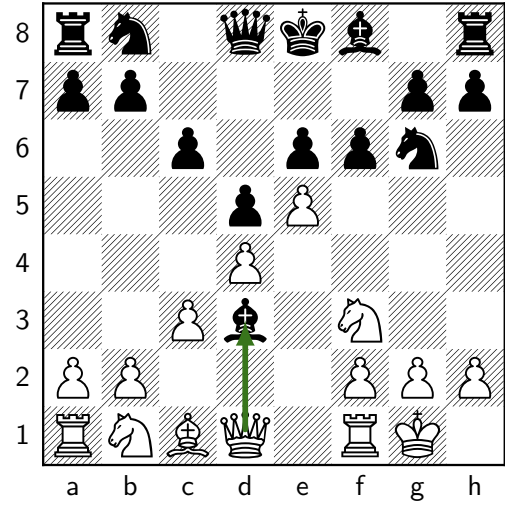
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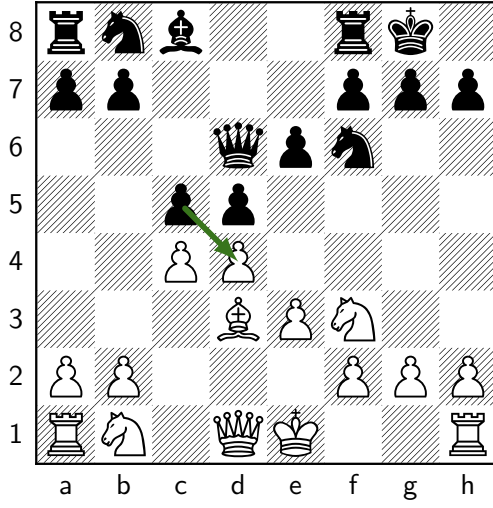


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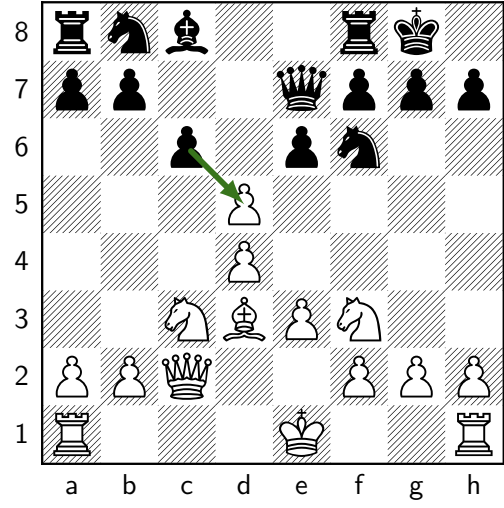


(d)

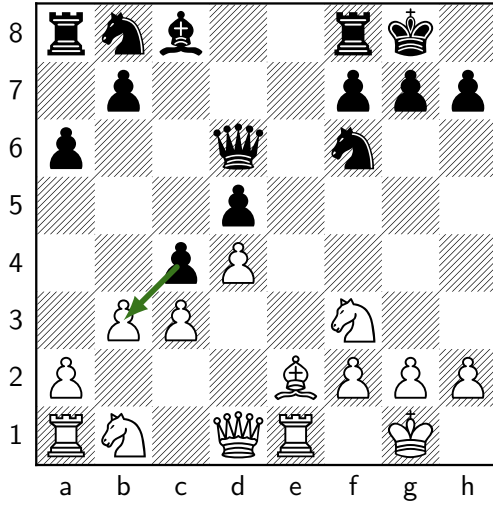
Figure 1: **Move 7, Black: Qxd6.** (a) Our position; Melanie plays Qxd6 (d8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#888` (ply 20) [structural distance  $d=71$ , QAP obj=132]: the master played Nxd3; under  $\mu$  our Qd1 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our bd3 as their Pa2, our Nf6 as their Bb2; transferred move d8d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#795` (ply 16) [structural distance  $d=77$ , QAP obj=106]: the master played Qxd3; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our nf3 as their Ra1, our Ra8 as their Bc1; transferred move d8d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#796` (ply 16) [structural distance  $d=77$ , QAP obj=106]: the master played Qxd3; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our nf3 as their Ra1, our Ra8 as their Bc1; transferred move d8d6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qxd6 (summed QAP objective 473, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



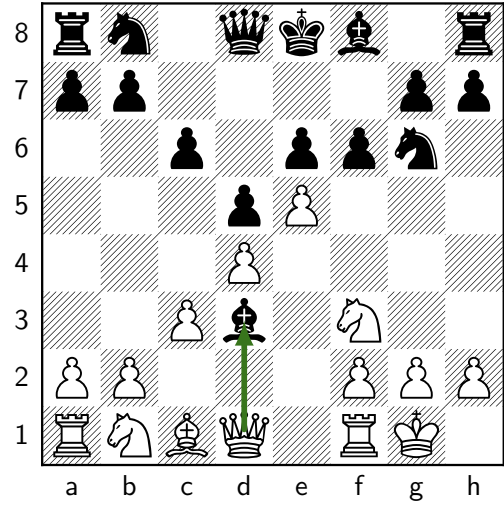
(a)



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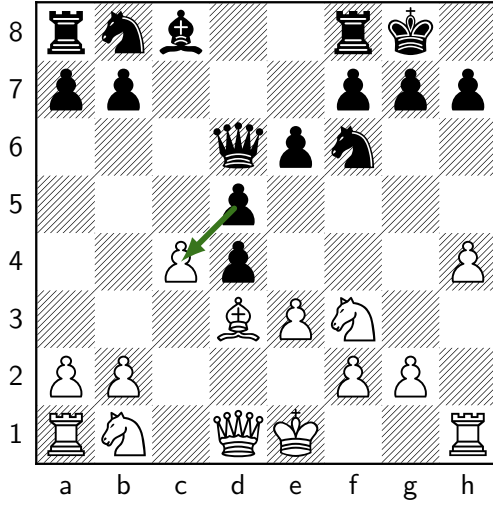


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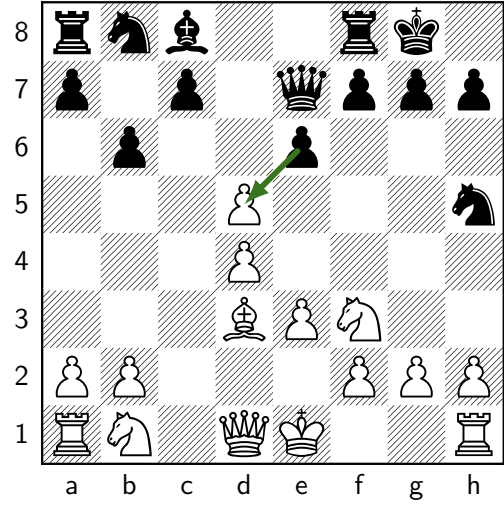


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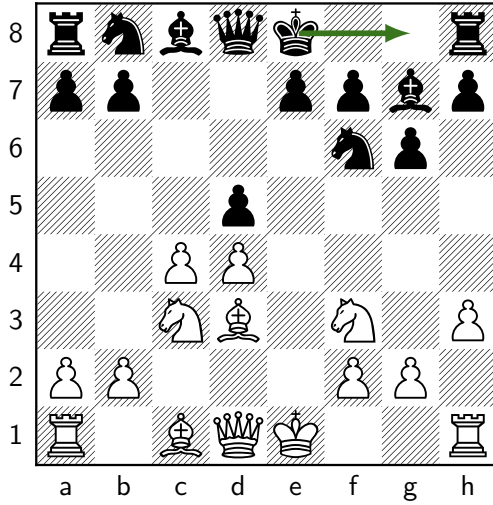
Figure 2: **Move 8, Black: cxd4.** (a) Our position; Melanie plays cxd4 (c5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#17 (ply 19) [structural distance  $d=64$ , QAP obj=94]: the master played cxd5; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Pc3$ ; cross-type roles: our  $Nb8$  as their  $Ra8$ , our  $pc4$  as their  $Nb8$ ; transferred move c5d4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#1337 (ply 21) [structural distance  $d=77$ , QAP obj=57]: the master played cxb3; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Pc5$ ; cross-type roles: our  $Nb8$  as their  $Ra8$ , our  $Bc8$  as their  $Nb8$ ; transferred move c5d4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#795 (ply 16) [structural distance  $d=86$ , QAP obj=31]: the master played Qxd3; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Qd1$  (a **cross-type** correspondence); cross-type roles: our  $Pe6$  as their  $Ra1$ , our  $Pc5$  as their  $Nb1$ ; transferred move c5d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to cxd4 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



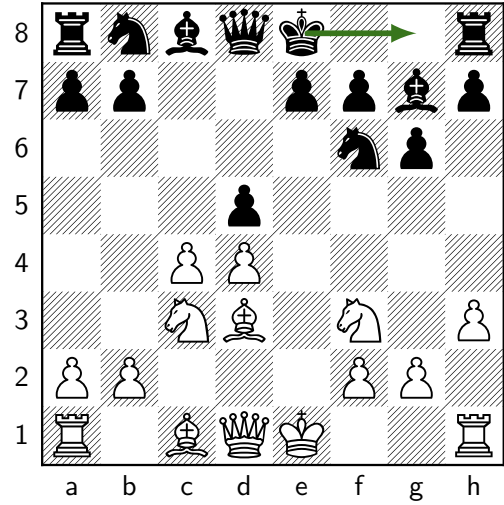
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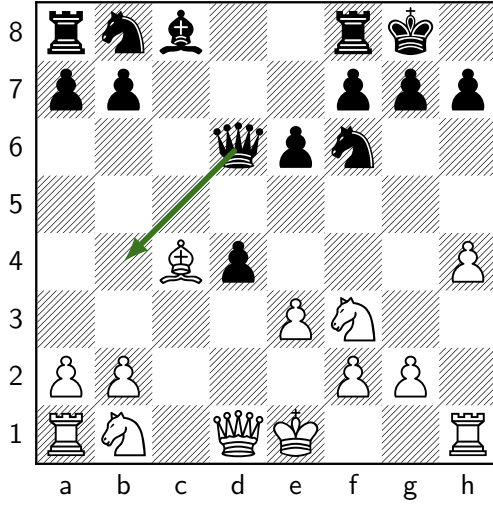


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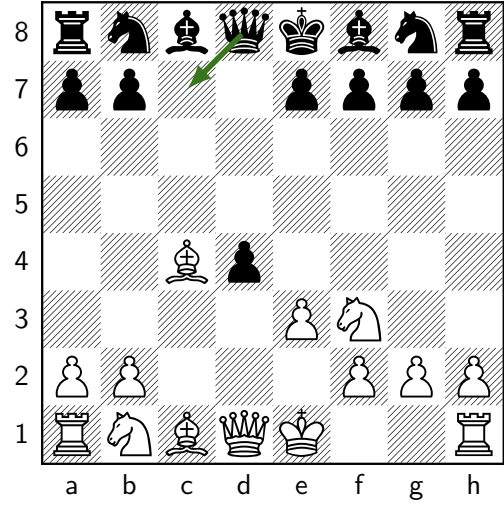


(d)

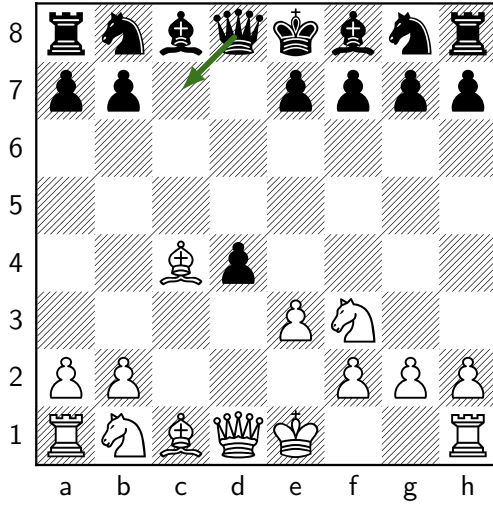
Figure 3: **Move 9, Black: dxc4.** (a) Our position; Melanie plays dxc4 (d5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3229` (ply 17) [structural distance  $d=60$ , QAP obj=53]: the master played `exd5`; under  $\mu$  our *Pd4* plays the role of that game’s *Pe3*; cross-type roles: our *pf2* as their *Nb8*, our *Nf6* as their *Bc8*; transferred move `d5c4` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3244` (ply 17) [structural distance  $d=90$ , QAP obj=85]: the master played `0-0`; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Ra8* as their *Nb8*, our *Rf8* as their *Bc8*; transferred move `g8h8`. (d) Retrieved from `ccr1_top_p0002#3292` (ply 17) [structural distance  $d=90$ , QAP obj=85]: the master played `0-0`; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Ra8* as their *Nb8*, our *Rf8* as their *Bc8*; transferred move `g8h8`. Synthesis: 1 retrieved analog transferred to dxc4 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



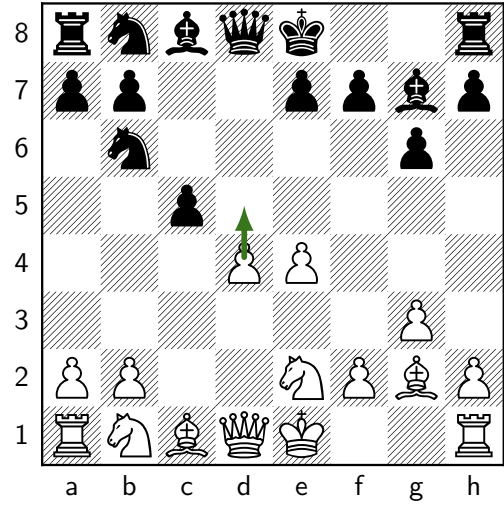
(a)



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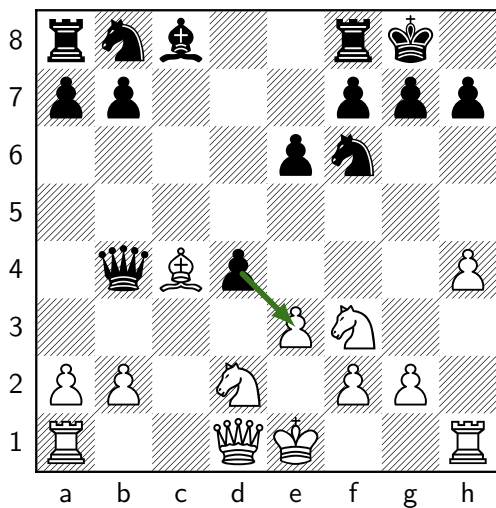


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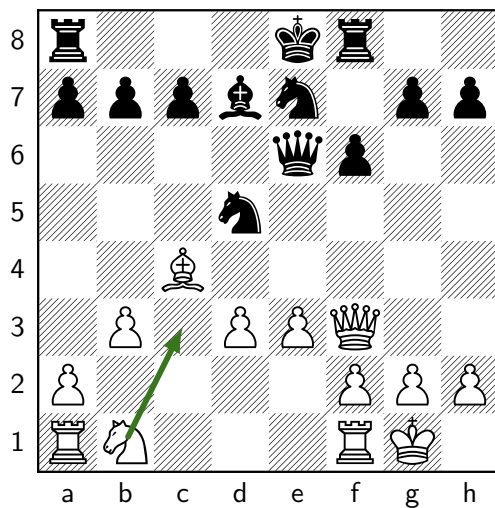


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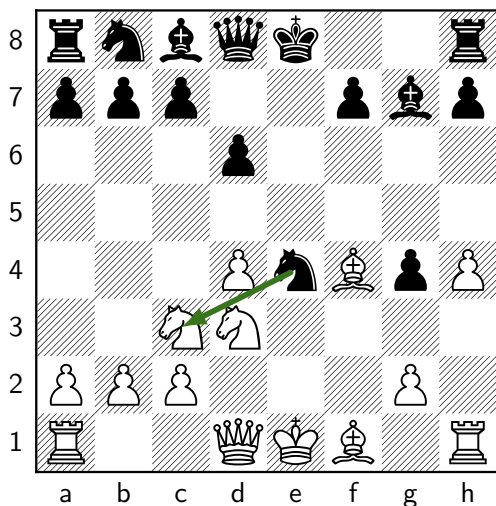
Figure 4: **Move 10, Black: Qb4+.** (a) Our position; Melanie plays Qb4+ (d6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1061` (ply 9) [structural distance  $d=74$ , QAP obj=171]: the master played Qc7; under  $\mu$  our Nf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra8, our Rf8 as their Nb8; transferred move f6h5. (c) Retrieved from `ccr1_top_p0002#1062` (ply 9) [structural distance  $d=74$ , QAP obj=171]: the master played Qc7; under  $\mu$  our Nf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra8, our Rf8 as their Nb8; transferred move f6h5. (d) Retrieved from `ccr1_top_p0000#2916` (ply 14) [structural distance  $d=80$ , QAP obj=110]: the master played d5; under  $\mu$  our Pd5 plays the role of that game’s Pd4; cross-type roles: our Nf6 as their Bc1, our Pf7 as their Rh1; transferred move d4d3. Synthesis: no retrieved analog transferred to Qb4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



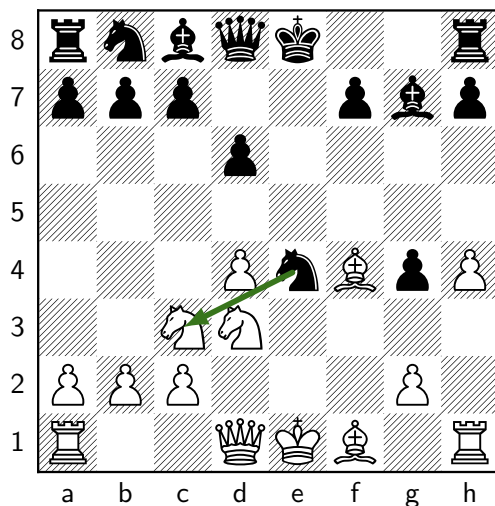
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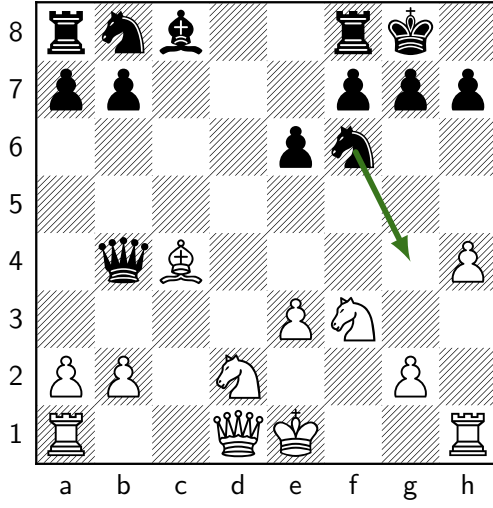


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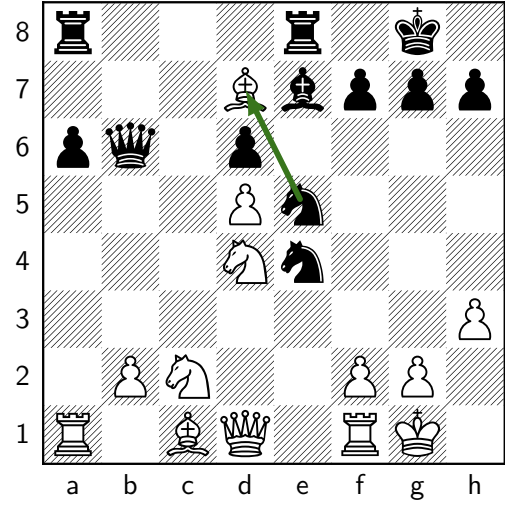


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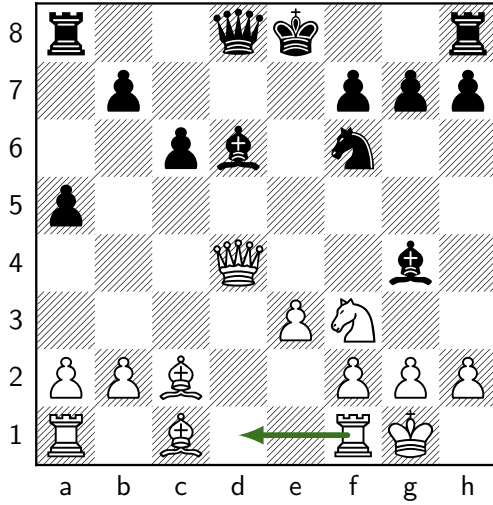
Figure 5: **Move 11, Black: dxe3**. (a) Our position; Melanie plays dxe3 (d4e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3796` (ply 26) [structural distance  $d=95$ , QAP obj=51]: the master played Nc3; under  $\mu$  our Pd5 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ph2, our Qb4 as their Bc4; transferred move d4e3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2287` (ply 17) [structural distance  $d=90$ , QAP obj=20]: the master played Nxc3; under  $\mu$  our Pd5 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Nb8, our Pf7 as their Bc8; transferred move d4e3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2549` (ply 17) [structural distance  $d=90$ , QAP obj=20]: the master played Nxc3; under  $\mu$  our Pd5 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Nb8, our Pf7 as their Bc8; transferred move d4e3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to dxe3 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



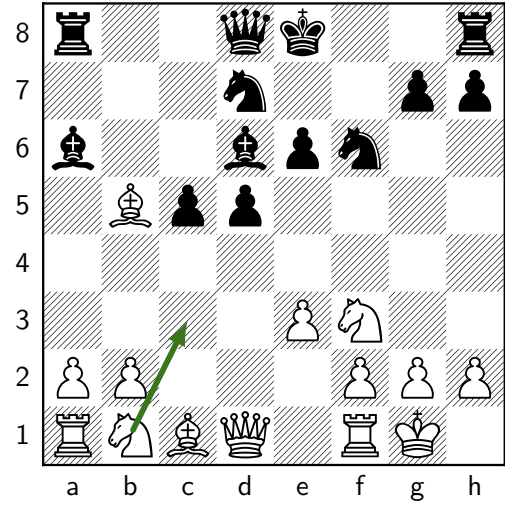
(a)



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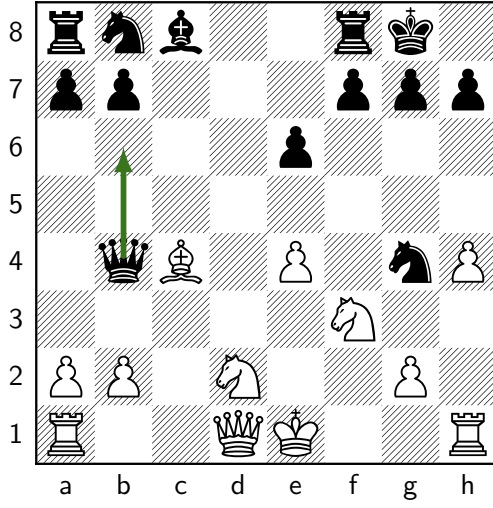


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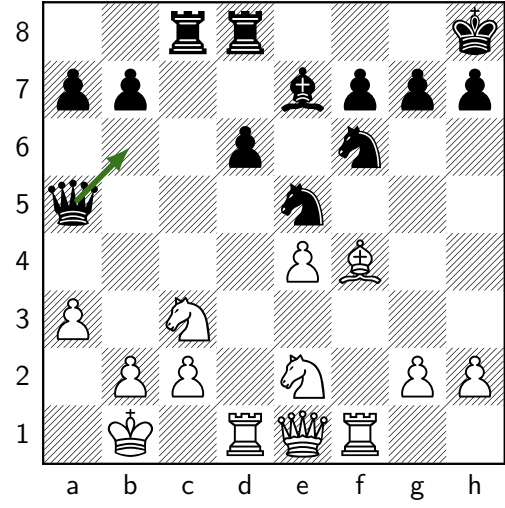


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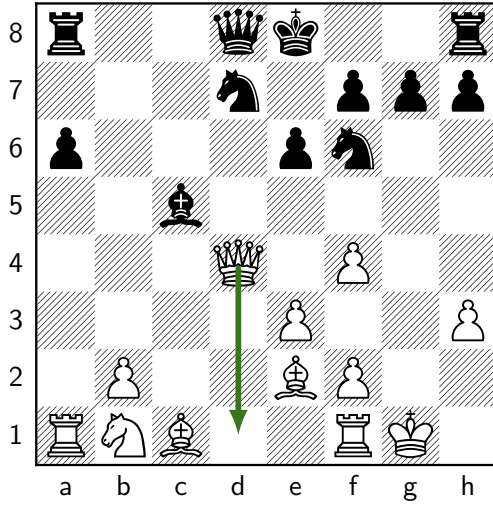
Figure 6: **Move 12, Black: Ng4**. (a) Our position; Melanie plays Ng4 (f6g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3547` (ply 37) [structural distance  $d=97$ , QAP obj=18]: the master played Nxd7; under  $\mu$  our Nf3 plays the role of that game’s Ne4; cross-type roles: our qd1 as their Ra8, our bc4 as their Re8; transferred move f6g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1869` (ply 26) [structural distance  $d=103$ , QAP obj=79]: the master played Rd1; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Nb8 as their Ra1, our Pf7 as their Bc1; transferred move f8e8. (d) Retrieved from `ccr1_top_p0002#3724` (ply 20) [structural distance  $d=56$ , QAP obj=61]: the master played Nc3; under  $\mu$  our Bc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra1, our Nf6 as their Pe3; transferred move c8d7. Synthesis: 1 retrieved analog transferred to Ng4 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



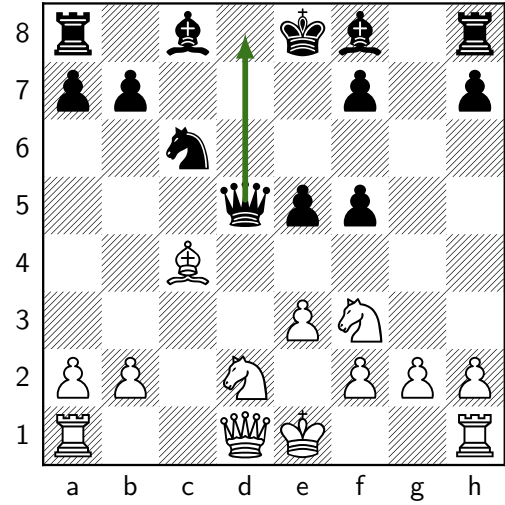
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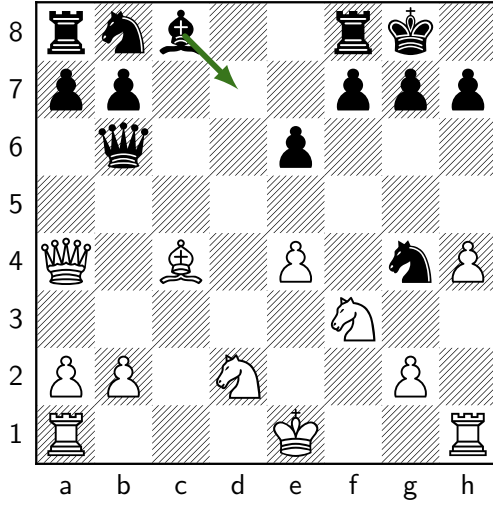
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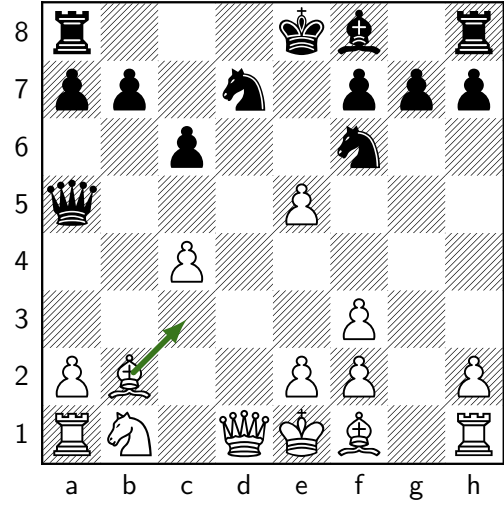
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Figure 7: **Move 13, Black: Qb6.** (a) Our position; Melanie plays Qb6 (b4b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#795` (ply 35) [structural distance  $d=106$ , QAP obj=57]: the master played Qb6; under  $\mu$  our Qb5 plays the role of that game’s Qa4; cross-type roles: our nf3 as their Rc8, our Ra8 as their Pb7; transferred move b4b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1283` (ply 28) [structural distance  $d=53$ , QAP obj=50]: the master played Qd1; under  $\mu$  our Qb5 plays the role of that game’s Qd4; cross-type roles: our Bc8 as their Ra1, our Ra8 as their Nb1; transferred move b4b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1207` (ply 21) [structural distance  $d=69$ , QAP obj=47]: the master played Qd8; under  $\mu$  our Qb5 plays the role of that game’s Qd4; cross-type roles: our Nb8 as their Ra8, our Pe6 as their Bc8; transferred move b4b6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qb6 (summed QAP objective 390, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

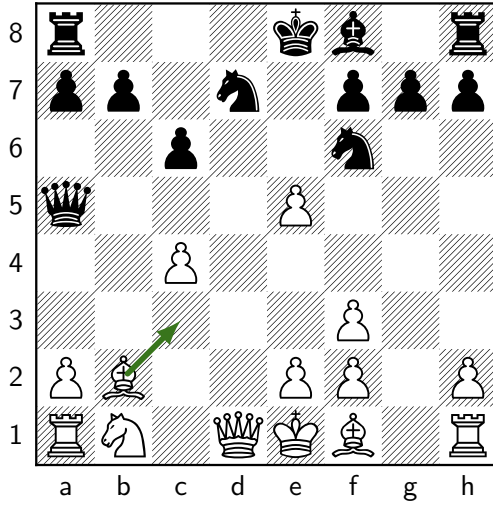




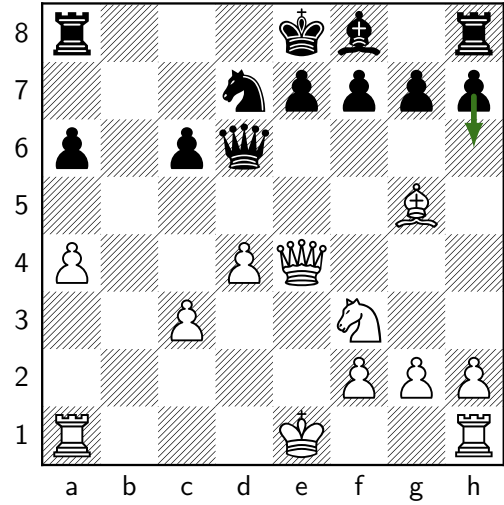
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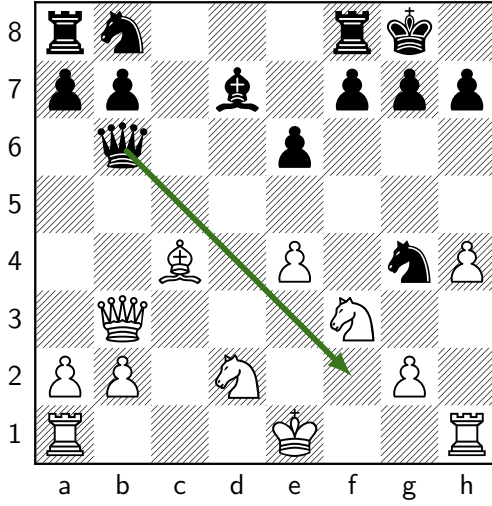


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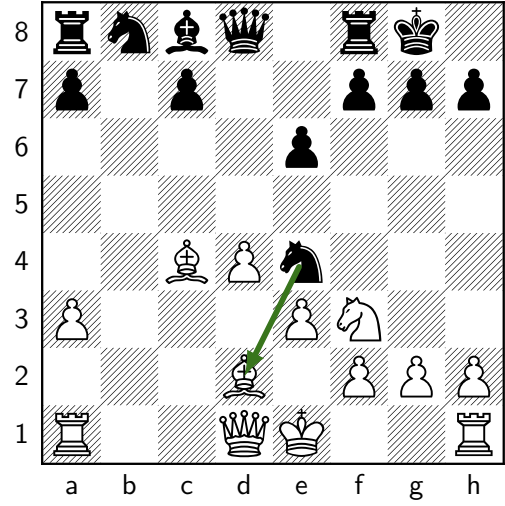


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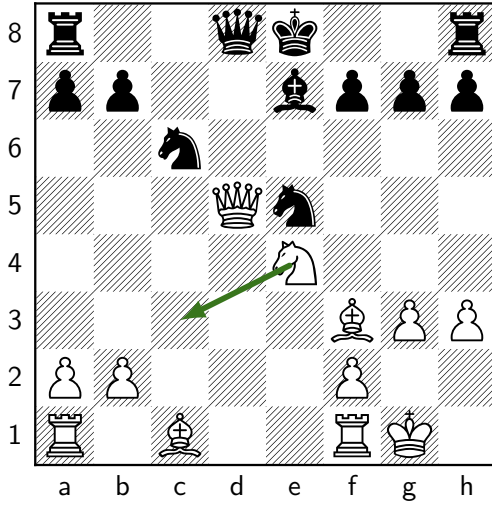
Figure 8: **Move 14, Black: Bd7.** (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#612 (ply 18) [structural distance  $d=88$ , QAP obj=25]: the master played Bc3; under  $\mu$  our Bc1 plays the role of that game’s Bb2; cross-type roles: our Rf8 as their Qd1, our Nb8 as their Rh1; transferred move c8d7 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#613 (ply 18) [structural distance  $d=88$ , QAP obj=25]: the master played Bc3; under  $\mu$  our Bc1 plays the role of that game’s Bb2; cross-type roles: our Rf8 as their Qd1, our Nb8 as their Rh1; transferred move c8d7 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#1562 (ply 27) [structural distance  $d=108$ , QAP obj=94]: the master played h6; under  $\mu$  our Pa2 plays the role of that game’s Ph2; cross-type roles: our Ra8 as their Bf8, our Pf7 as their Rh8; transferred move a7a5. Synthesis: 2 retrieved analogs transferred to Bd7 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



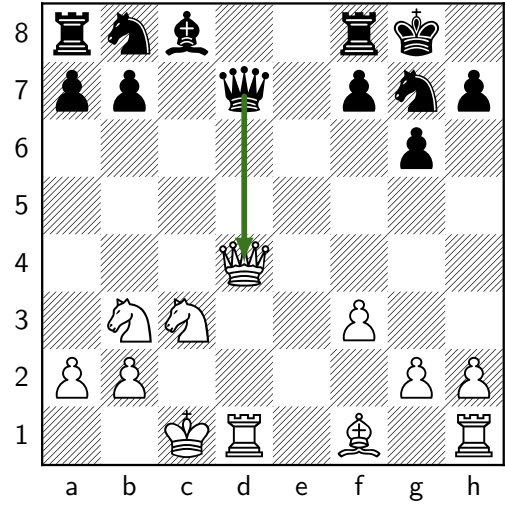
(a)



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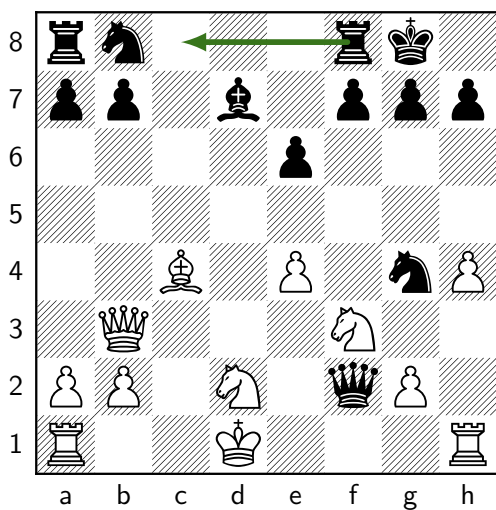


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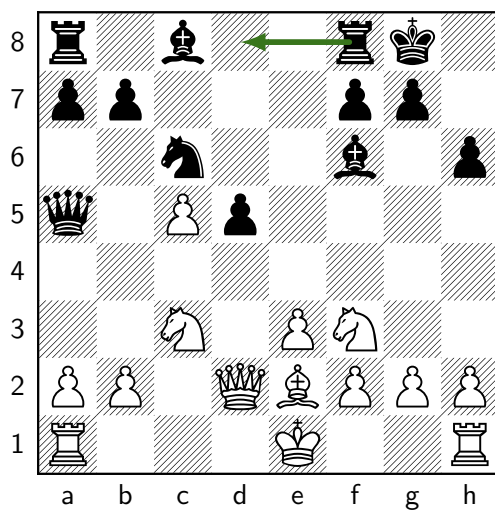


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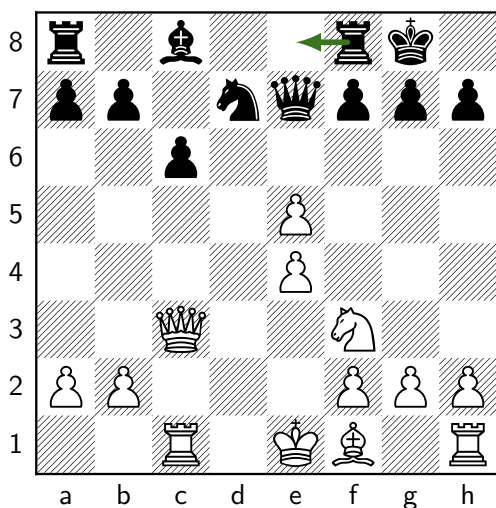
Figure 9: **Move 15, Black: Qf2+.** (a) Our position; Melanie plays Qf2+ (b6f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#429` (ply 21) [structural distance  $d=104$ , QAP obj=15]: the master played Nxd2; under  $\mu$  our Qb3 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our bc4 as their Ra8, our Pe6 as their Nb8; transferred move b6f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3464` (ply 28) [structural distance  $d=101$ , QAP obj=81]: the master played Nc3; under  $\mu$  our Pe3 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Bd7 as their Ra1, our rh1 as their Bc1; transferred move e6e5. (d) Retrieved from `ccr1_top_p0001#411` (ply 33) [structural distance  $d=97$ , QAP obj=52]: the master played Qxd4; under  $\mu$  our Qb3 plays the role of that game’s Qd2; cross-type roles: our Pb7 as their Ra8, our Ra8 as their Nb8; transferred move b6b3. Synthesis: 1 retrieved analog transferred to Qf2+ (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



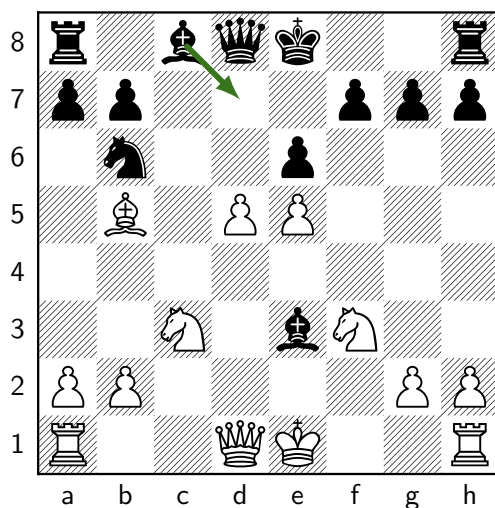
(a)



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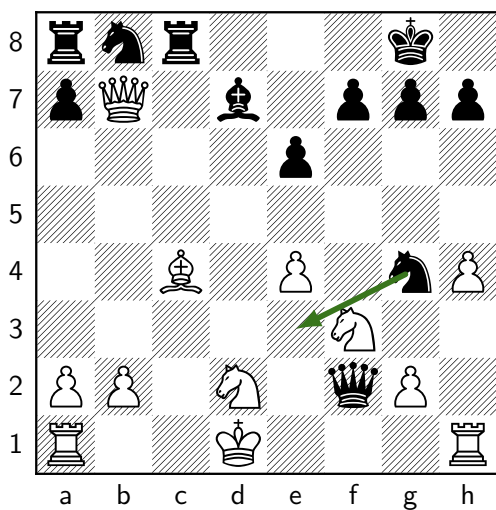


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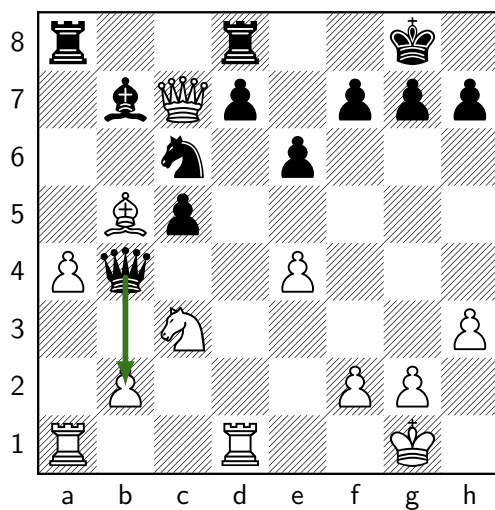


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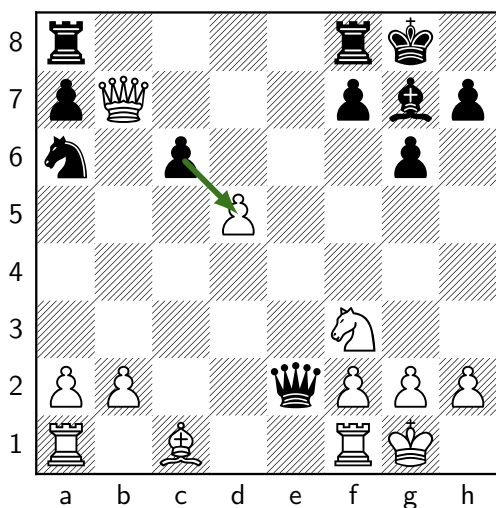
Figure 10: **Move 16, Black: Rc8 (f8c8)**. (a) Our position; Melanie plays Rc8 (f8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#228` (ply 23) [structural distance  $d=107$ , QAP obj=29]: the master played Rd8; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pf7 as their Bc8, our Nb8 as their Pg7; transferred move f8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1183` (ply 23) [structural distance  $d=115$ , QAP obj=26]: the master played Re8; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Nb8 as their Ra8, our Pf7 as their Bc8; transferred move f8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#729` (ply 23) [structural distance  $d=115$ , QAP obj=20]: the master played Bd7; under  $\mu$  our Rf1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Qf2 as their Ra8, our Ng4 as their Rh8; transferred move f8c8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rc8 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



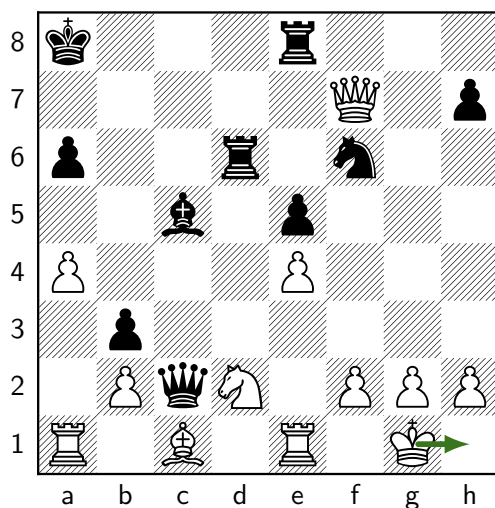
(a)



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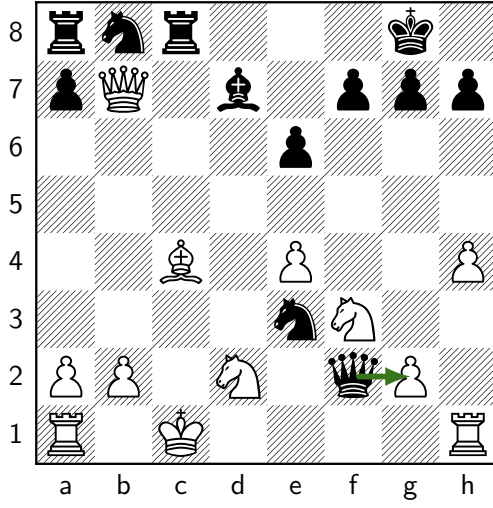


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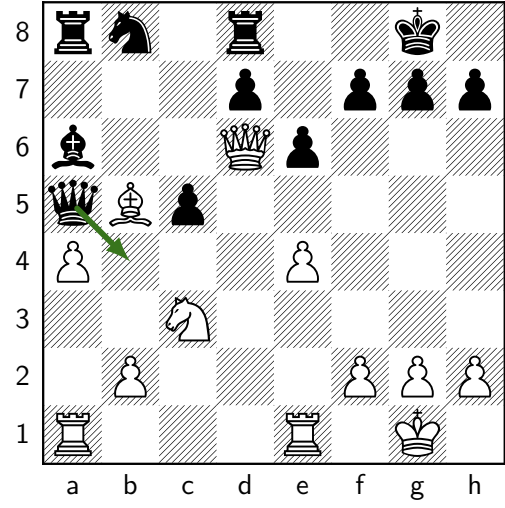


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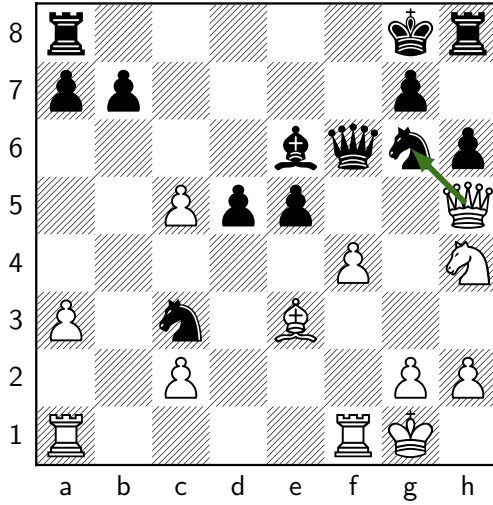
Figure 11: **Move 17, Black: Ne3+.** (a) Our position; Melanie plays Ne3+ (g4e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1056` (ply 37) [structural distance  $d=128$ , QAP obj=123]: the master played Qxb2; under  $\mu$  our Qf7 plays the role of that game’s Qb5; cross-type roles: our Bd7 as their Ra8, our Rc8 as their Bb7; transferred move f2g2. (c) Retrieved from `ccr1_top_p0002#2162` (ply 33) [structural distance  $d=131$ , QAP obj=105]: the master played cxd5; under  $\mu$  our Bd2 plays the role of that game’s Pc3 (a **cross-type** correspondence); cross-type roles: our nd2 as their Rf8, our Nb8 as their Pa7; transferred move d7e8. (d) Retrieved from `ccr1_top_p0003#2252` (ply 44) [structural distance  $d=112$ , QAP obj=95]: the master played Kh1; under  $\mu$  our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pe6 as their Ra1, our Ra8 as their Bc1; transferred move g8h8. Synthesis: no retrieved analog transferred to Ne3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



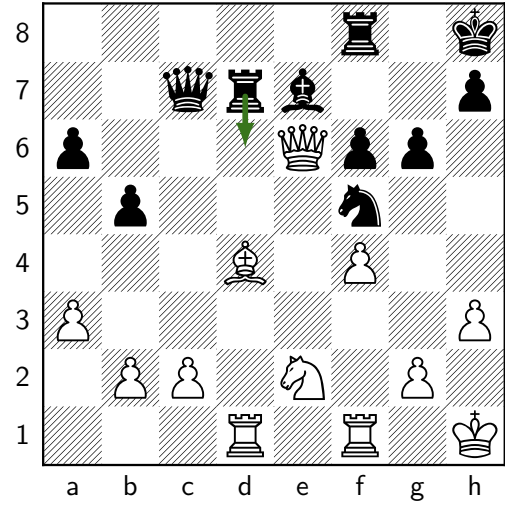
(a)



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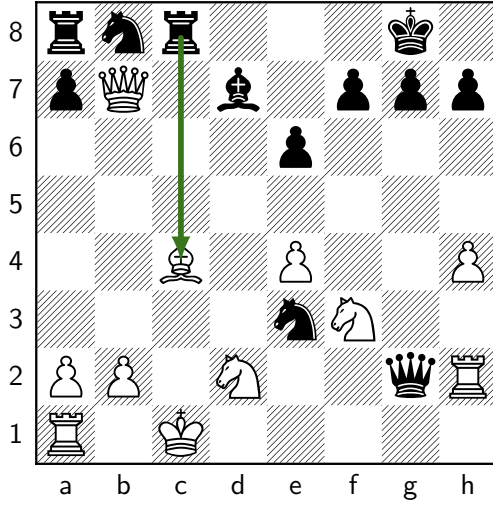


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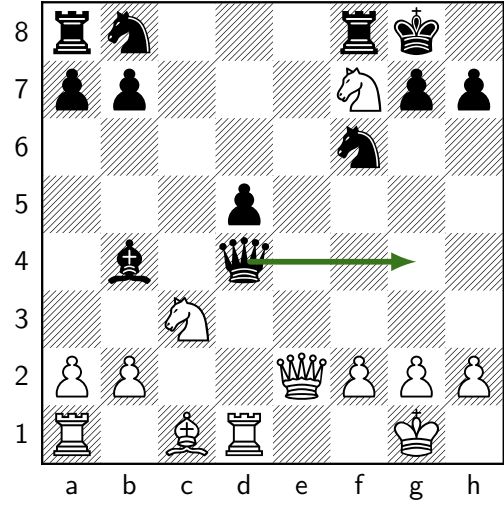


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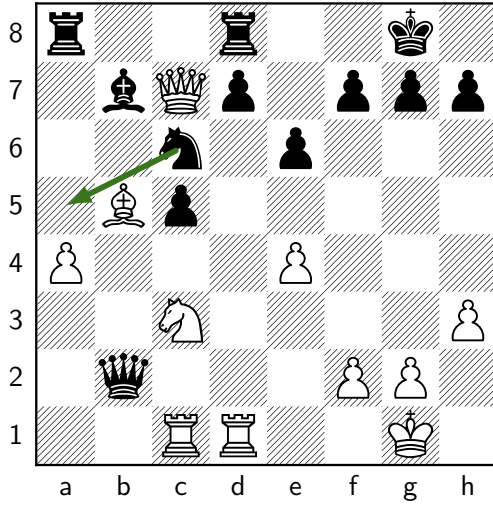
Figure 12: **Move 18, Black: Qxg2.** (a) Our position; Melanie plays Qxg2 (f2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1056` (ply 31) [structural distance  $d=122$ , QAP obj=54]: the master played Qb4; under  $\mu$  our Qf7 plays the role of that game’s Qa4; cross-type roles: our bc4 as their Ra8, our Bd7 as their Pd7; transferred move f2g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1758` (ply 36) [structural distance  $d=147$ , QAP obj=50]: the master played Qxg6; under  $\mu$  our Qf7 plays the role of that game’s Qh5; cross-type roles: our qb7 as their Ra1, our pb2 as their Rf1; transferred move f2g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1333` (ply 51) [structural distance  $d=141$ , QAP obj=144]: the master played Rd6; under  $\mu$  our Rc1 plays the role of that game’s Rd2; cross-type roles: our ph4 as their Rf8, our Bd7 as their Qc7; transferred move c8c6. Synthesis: 2 retrieved analogs transferred to Qxg2 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



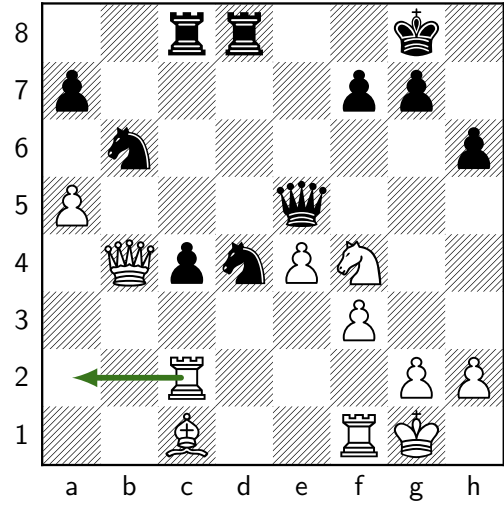
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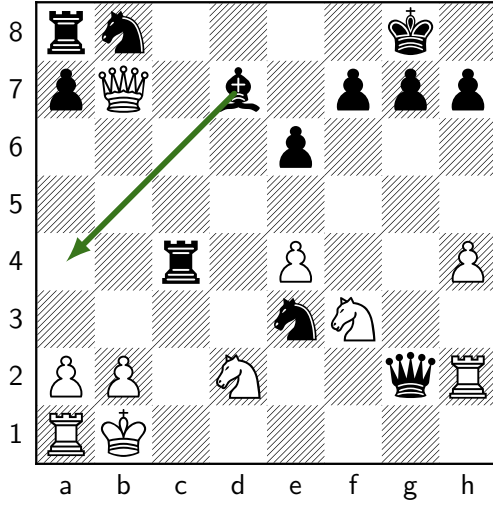


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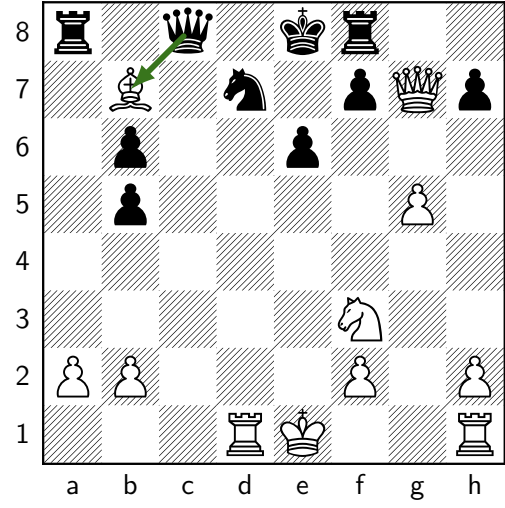


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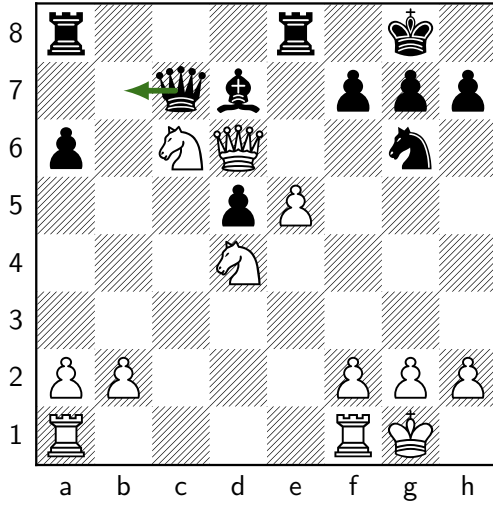
Figure 13: **Move 19, Black: Rxc4+.** (a) Our position; Melanie plays Rxc4+ (c8c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2361` (ply 27) [structural distance  $d=154$ , QAP obj=37]: the master played Qg4; under  $\mu$  our *Rc1* plays the role of that game’s *Qd5* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra8*, our *pa2* as their *Nb8*; transferred move c8c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1056` (ply 39) [structural distance  $d=131$ , QAP obj=71]: the master played Na5; under  $\mu$  our *Nb1* plays the role of that game’s *Nc3*; cross-type roles: our *Bd7* as their *Ra8*, our *Rc8* as their *Bb7*; transferred move b8c6. (d) Retrieved from `ccr1_top_p0002#2719` (ply 50) [structural distance  $d=156$ , QAP obj=57]: the master played Ra2; under  $\mu$  our *Rc1* plays the role of that game’s *Rc2*; cross-type roles: our *Ra8* as their *Bc1*, our *Bd7* as their *Rf1*; transferred move c8e8. Synthesis: 1 retrieved analog transferred to Rxc4+ (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



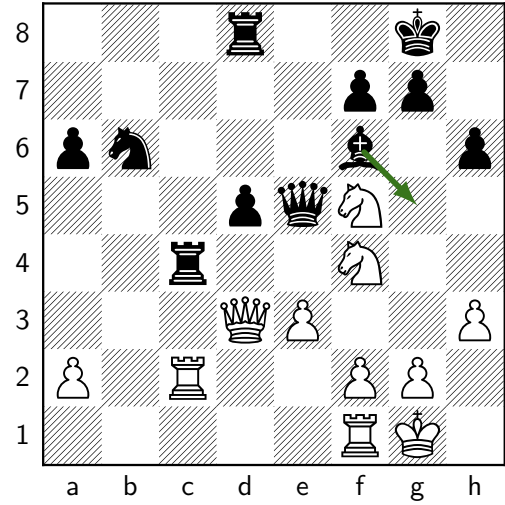
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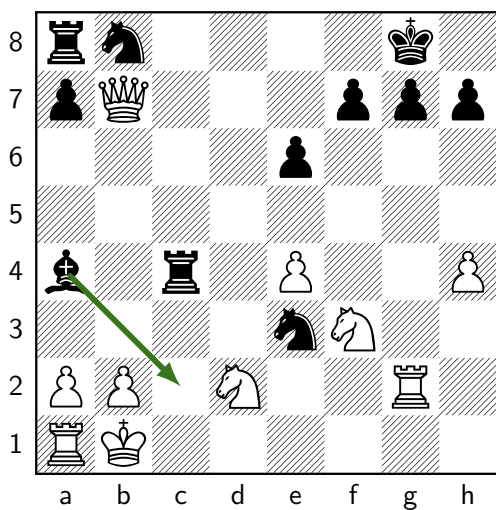
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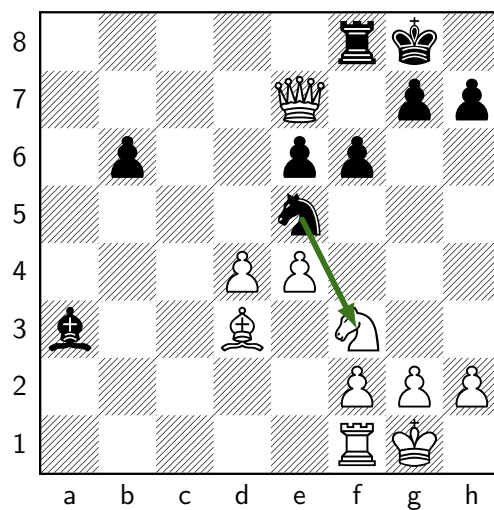
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Figure 14: **Move 20, Black: Ba4.** (a) Our position; Melanie plays Ba4 (d7a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2567` (ply 37) [structural distance  $d=142$ , QAP obj=54]: the master played Qxb7; under  $\mu$  our Qg7 plays the role of that game’s Qc1; cross-type roles: our Qg2 as their Ra8, our rh2 as their bb7; transferred move g2f3. (c) Retrieved from `ccr1_top_p0000#2203` (ply 43) [structural distance  $d=136$ , QAP obj=49]: the master played Qb7; under  $\mu$  our Qg7 plays the role of that game’s Qc2; cross-type roles: our Pg7 as their Ra8, our Ph7 as their Re8; transferred move g2g3. (d) Retrieved from `ccr1_top_p0003#2026` (ply 47) [structural distance  $d=133$ , QAP obj=44]: the master played Bg5; under  $\mu$  our Ph2 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our ph4 as their Rd8, our Bd7 as their Pf7; transferred move h7h6. Synthesis: no retrieved analog transferred to Ba4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

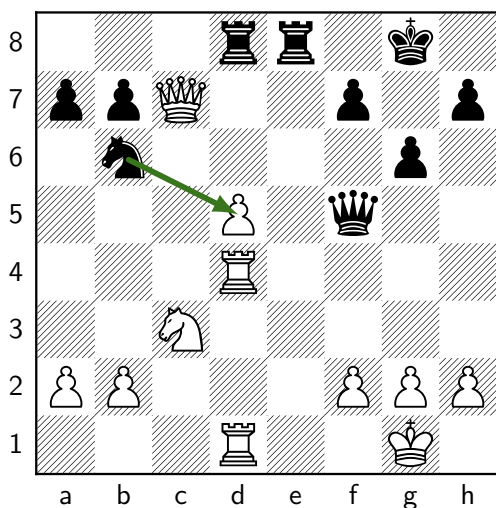




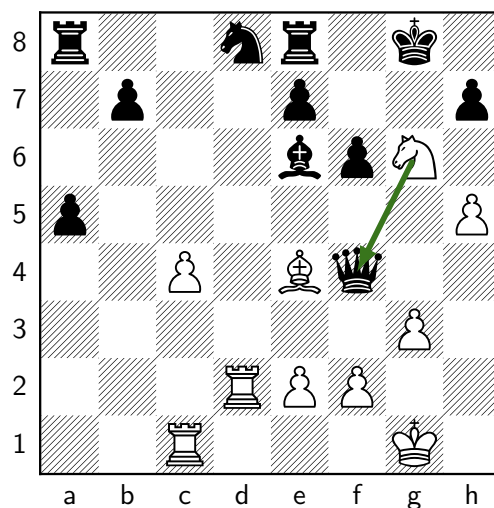
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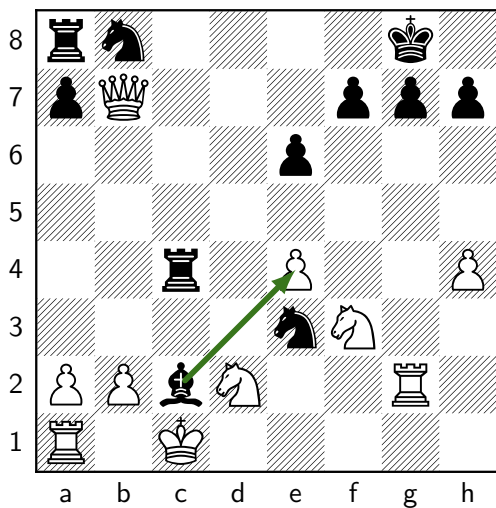
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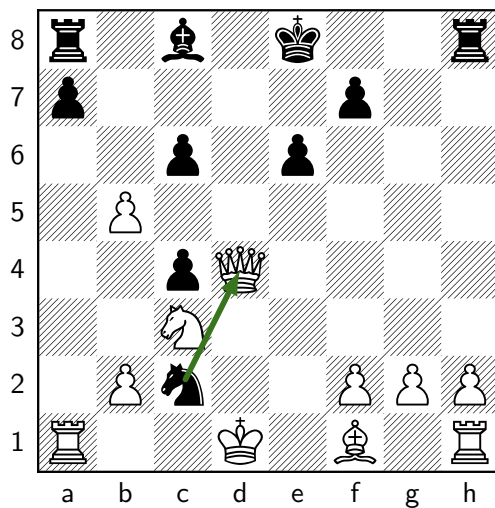
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Figure 15: **Move 21, Black: Bc2+.** (a) Our position; Melanie plays Bc2+ (a4c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3067` (ply 43) [structural distance  $d=105$ , QAP obj=61]: the master played Nxf3+; under  $\mu$  our Rc5 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pg7, our Ba4 as their Pb6; transferred move c4c3. (c) Retrieved from `ccr1_top_p0002#3820` (ply 43) [structural distance  $d=117$ , QAP obj=49]: the master played Nxd5; under  $\mu$  our Nb1 plays the role of that game’s Nb3; cross-type roles: our Ba4 as their Re8, our rg2 as their Pg6; transferred move b8c6. (d) Retrieved from `ccr1_top_p0003#2795` (ply 48) [structural distance  $d=118$ , QAP obj=44]: the master played Nxf4; under  $\mu$  our Nb1 plays the role of that game’s Ng6; cross-type roles: our Pf7 as their Rc1, our Rc4 as their Pc4; transferred move b8c6. Synthesis: no retrieved analog transferred to Bc2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

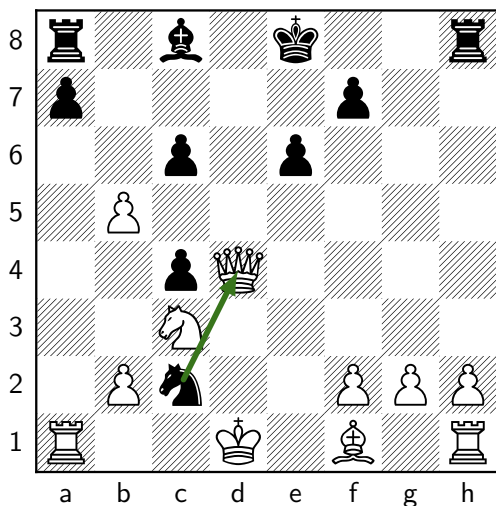




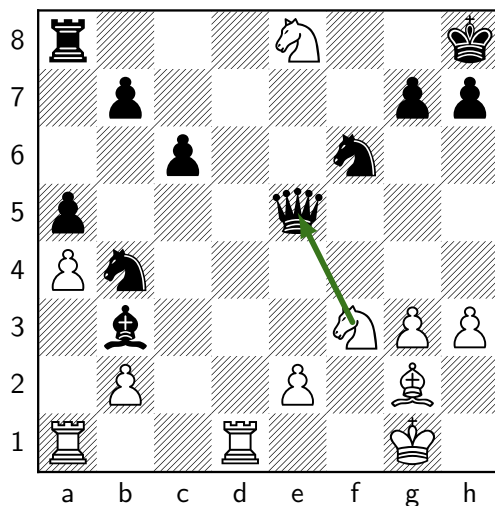
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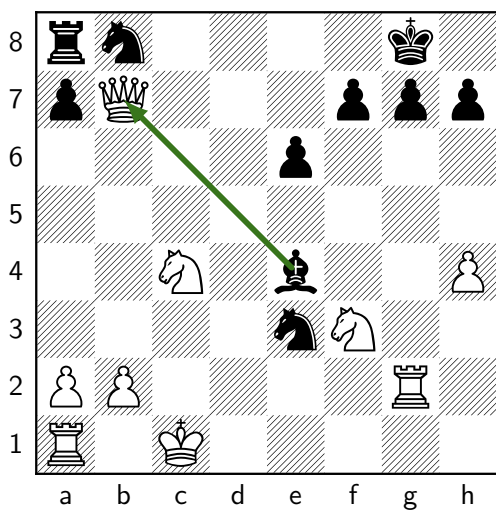


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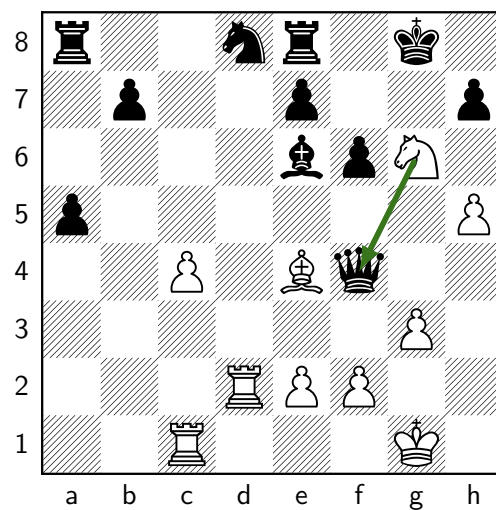


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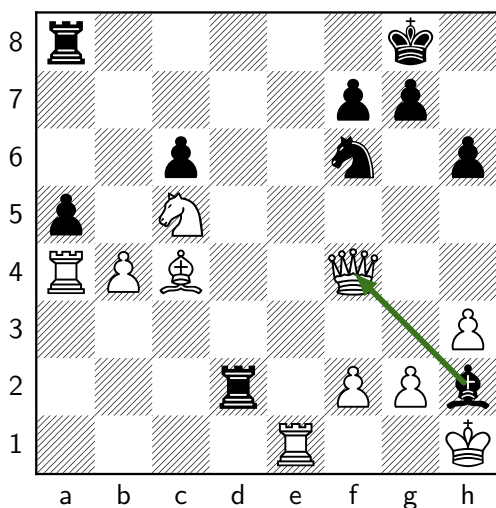
Figure 16: **Move 22, Black: Bxe4+.** (a) Our position; Melanie plays Bxe4+ (c2e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2868` (ply 33) [structural distance  $d=116$ , QAP obj=69]: the master played Nxd4; under  $\mu$  our Ne6 plays the role of that game’s Nc7; cross-type roles: our Ph7 as their Ra8, our Nb8 as their Bc8; transferred move e3g2. (c) Retrieved from `ccr1_top_p0002#3901` (ply 33) [structural distance  $d=116$ , QAP obj=69]: the master played Nxd4; under  $\mu$  our Ne6 plays the role of that game’s Nc7; cross-type roles: our Ph7 as their Ra8, our Nb8 as their Bc8; transferred move e3g2. (d) Retrieved from `ccr1_top_p0000#1625` (ply 58) [structural distance  $d=125$ , QAP obj=55]: the master played Nxe5; under  $\mu$  our Ne6 plays the role of that game’s Nf3; cross-type roles: our Bc2 as their Ra1, our Ph7 as their Bg2; transferred move e3g2. Synthesis: no retrieved analog transferred to Bxe4+ – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



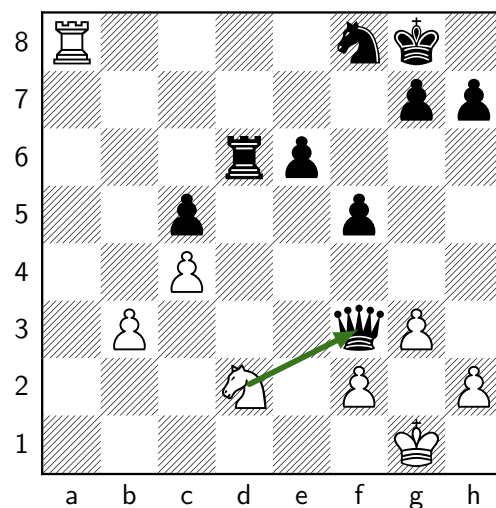
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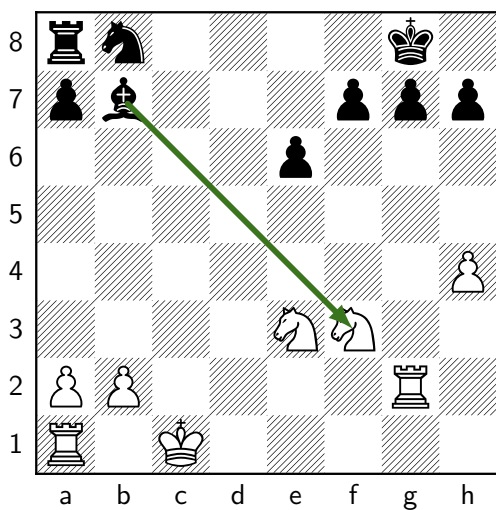


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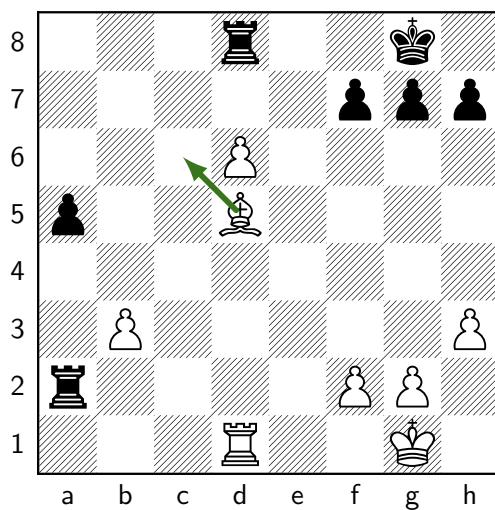


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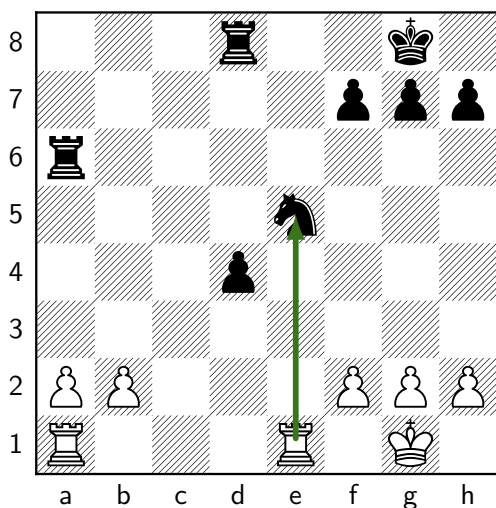
Figure 17: **Move 23, Black: Bxb7**. (a) Our position; Melanie plays Bxb7 (e4b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#2795` (ply 48) [structural distance  $d=107$ , QAP obj=34]: the master played Nxf4; under  $\mu$  our Be5 plays the role of that game’s Ng6 (a **cross-type** correspondence); cross-type roles: our Be4 as their Pg3, our Ne3 as their Pc4; transferred move e4b7 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#2143` (ply 53) [structural distance  $d=115$ , QAP obj=30]: the master played Bxf4; under  $\mu$  our Be5 plays the role of that game’s Bh7; cross-type roles: our nf3 as their Pa5, our Ph7 as their nc5; transferred move e4b7 (**endorses this move**). (d) Retrieved from `cctl_top_p0003#467` (ply 60) [structural distance  $d=112$ , QAP obj=29]: the master played Nxf3; under  $\mu$  our Be5 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pf2, our nc4 as their pe6; transferred move e4b7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxb7 (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



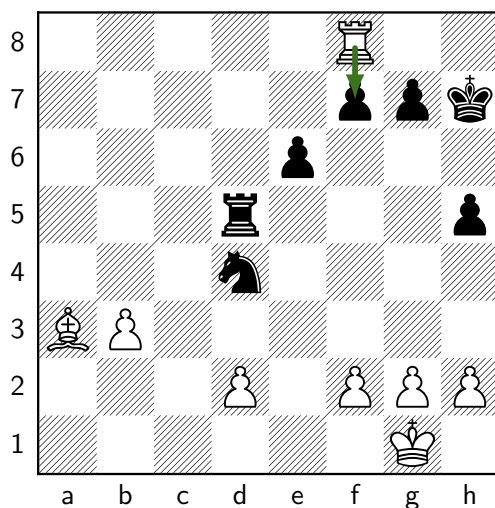
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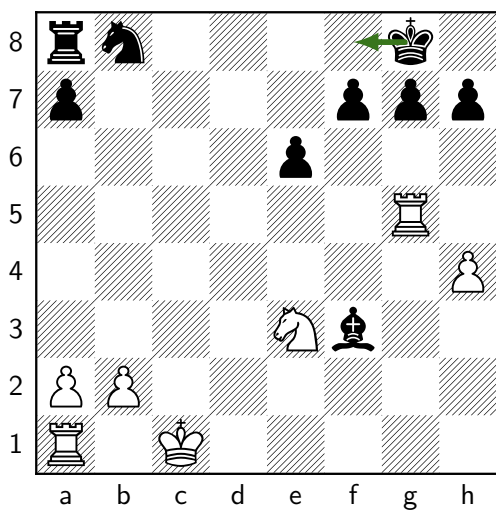


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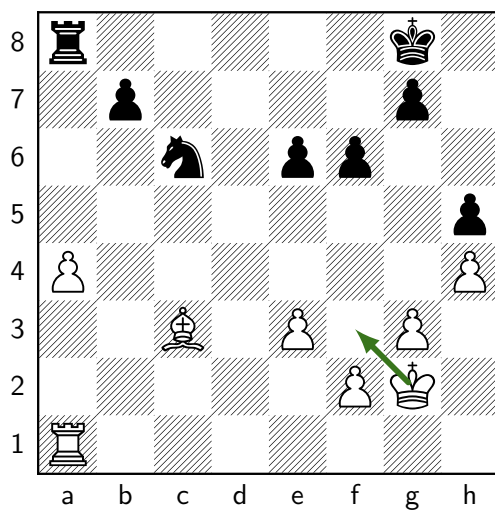


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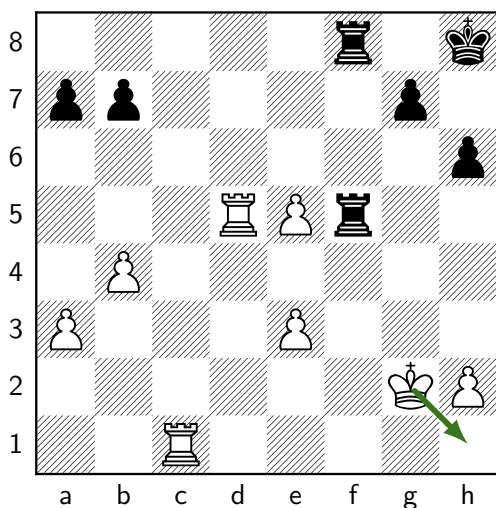
Figure 18: **Move 24, Black: Bxf3**. (a) Our position; Melanie plays Bxf3 (b7f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#196 (ply 58) [structural distance  $d=53$ , QAP obj=48]: the master played Bc6; under  $\mu$  our Bb2 plays the role of that game’s Bd5; cross-type roles: our ne3 as their Rd1, our Ra8 as their Pg2; transferred move b7f3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0003#117 (ply 42) [structural distance  $d=50$ , QAP obj=34]: the master played Rxe5; under  $\mu$  our Bb2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our ne3 as their pd4, our Nb8 as their ph7; transferred move b7f3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#80 (ply 54) [structural distance  $d=54$ , QAP obj=15]: the master played Rxf7; under  $\mu$  our Bb2 plays the role of that game’s Rf8 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pd2, our Pe6 as their Ba3; transferred move b7f3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bxf3 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



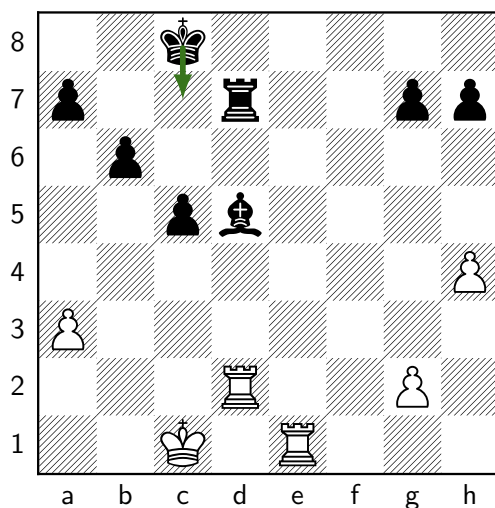
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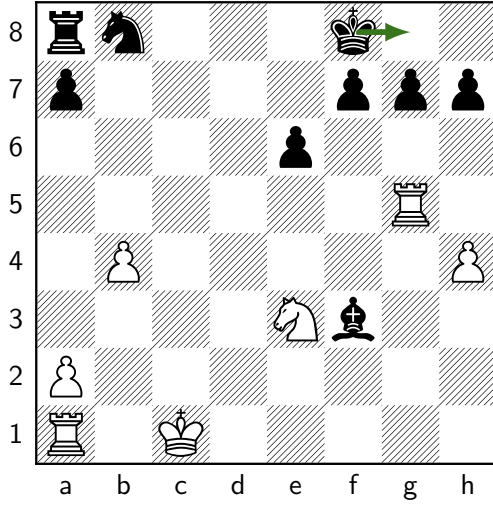


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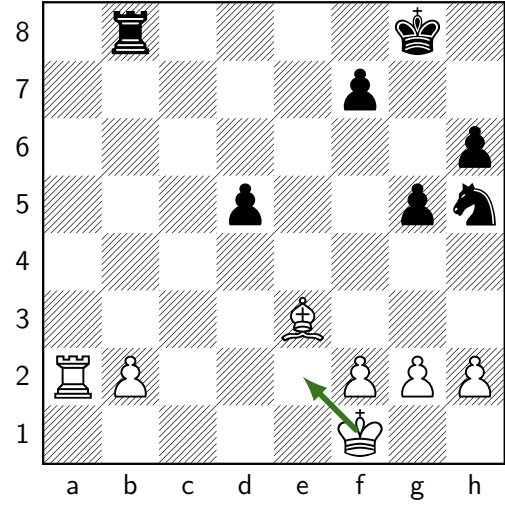


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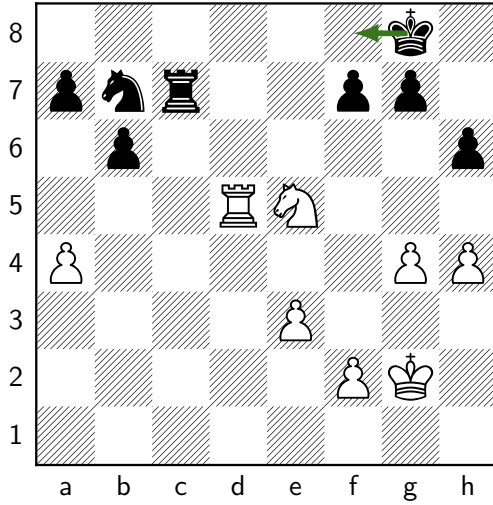
Figure 19: **Move 25, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1446` (ply 62) [structural distance  $d=42$ , QAP obj=24]: the master played Kf3; under  $\mu$  our Kg1 plays the role of that game’s Kg2; cross-type roles: our ph4 as their Ra1, our ne3 as their Pf2; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2234` (ply 64) [structural distance  $d=43$ , QAP obj=23]: the master played Kh1; under  $\mu$  our Kg1 plays the role of that game’s Kg2; cross-type roles: our ne3 as their Rc1, our Ra8 as their Ph2; transferred move g8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#452` (ply 53) [structural distance  $d=43$ , QAP obj=23]: the master played Kc7; under  $\mu$  our Kg1 plays the role of that game’s Kc1; cross-type roles: our Ph7 as their Rd7, our Ra8 as their Ph7; transferred move g8f8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kf8 (summed QAP objective 166, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



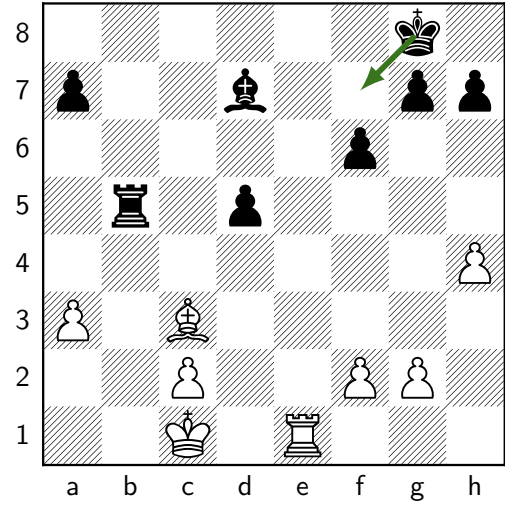
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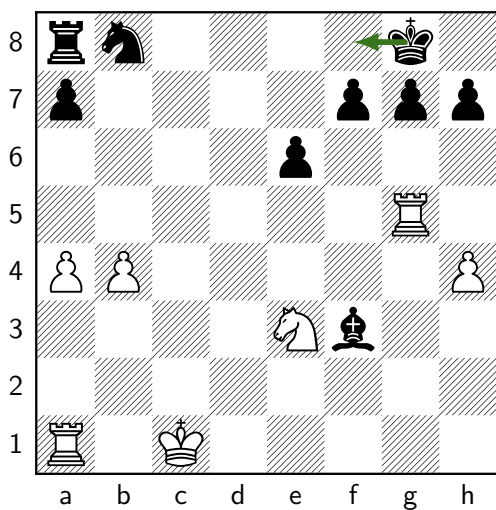


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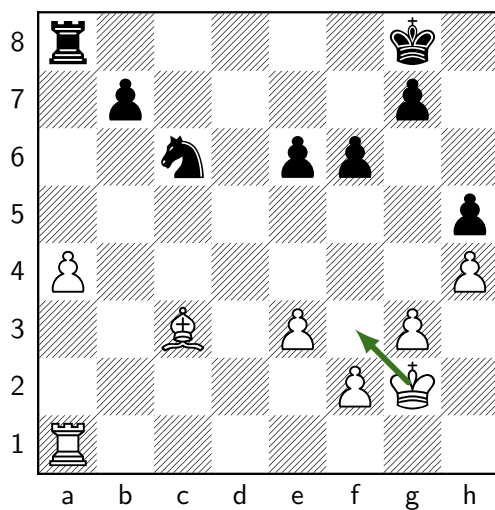


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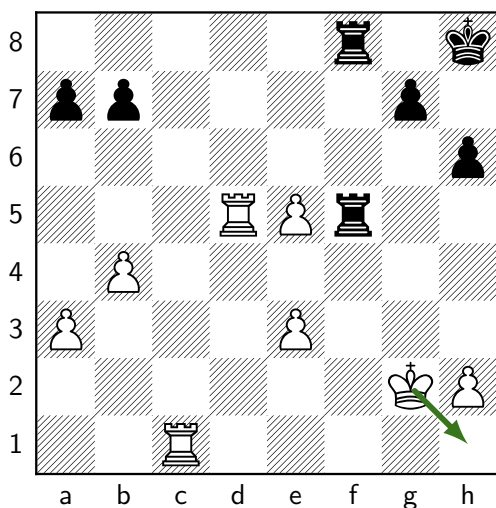
Figure 20: **Move 26, Black: Kg8**. (a) Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#1687` (ply 58) [structural distance  $d=45$ , QAP obj=22]: the master played Kc2; under  $\mu$  our Kf1 plays the role of that game’s Kf1; cross-type roles: our ne3 as their Ph2, our rg5 as their Be3; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#885` (ply 63) [structural distance  $d=46$ , QAP obj=21]: the master played Kf8; under  $\mu$  our Kf1 plays the role of that game’s Kg1; cross-type roles: our Ra8 as their Pa7, our Ph7 as their Nb7; transferred move f8g8 (**endorses this move**). (d) Retrieved from `ccl1_top_p0003#1971` (ply 55) [structural distance  $d=44$ , QAP obj=16]: the master played Kf7; under  $\mu$  our Kf1 plays the role of that game’s Kg1; cross-type roles: our Ra8 as their Bd7, our Bf3 as their Rb5; transferred move f8g8 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg8 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



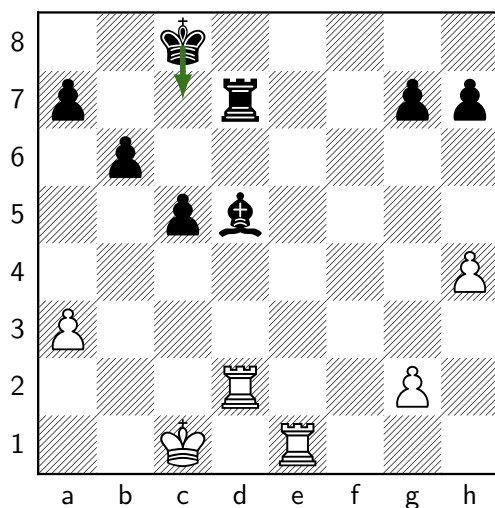
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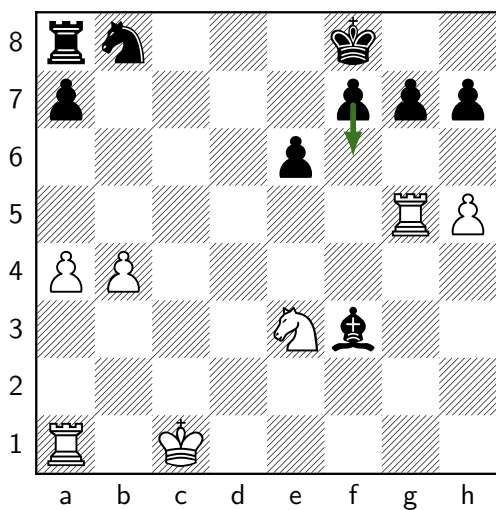


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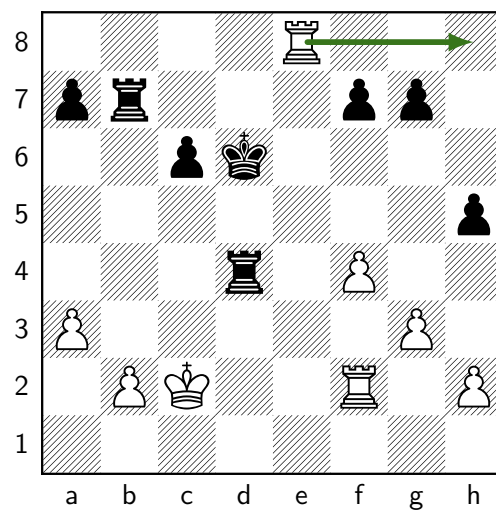


(d)

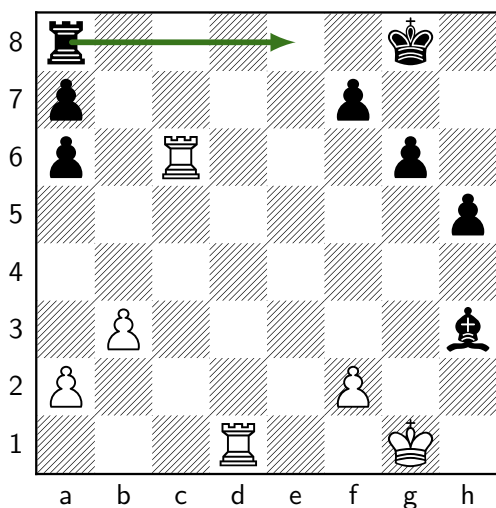
Figure 21: **Move 27, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1446` (ply 62) [structural distance  $d=46$ , QAP obj=24]: the master played Kf3; under  $\mu$  our Kg1 plays the role of that game’s Kg2; cross-type roles: our Pe6 as their Ra1, our rg5 as their Bc3; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2234` (ply 64) [structural distance  $d=45$ , QAP obj=23]: the master played Kh1; under  $\mu$  our Kg1 plays the role of that game’s Kg2; cross-type roles: our Nb8 as their Ph2, our ne3 as their Pb4; transferred move g8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#452` (ply 53) [structural distance  $d=44$ , QAP obj=23]: the master played Kc7; under  $\mu$  our Kg1 plays the role of that game’s Kc1; cross-type roles: our Ph7 as their Rd7, our Nb8 as their Ph7; transferred move g8f8 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Kf8 (summed QAP objective 240, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



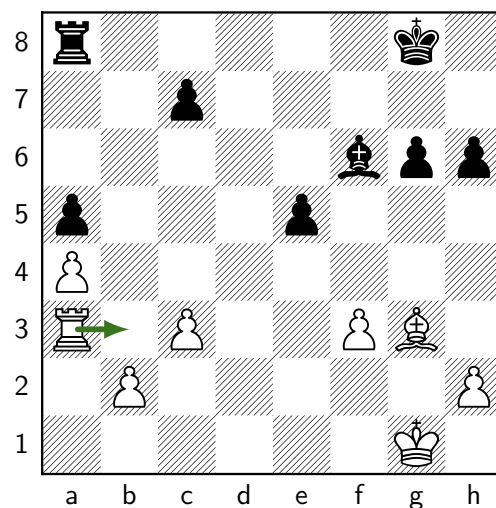
(a)



(b)

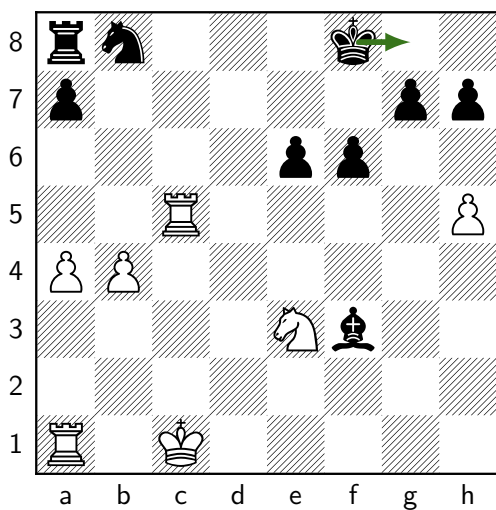


(c)

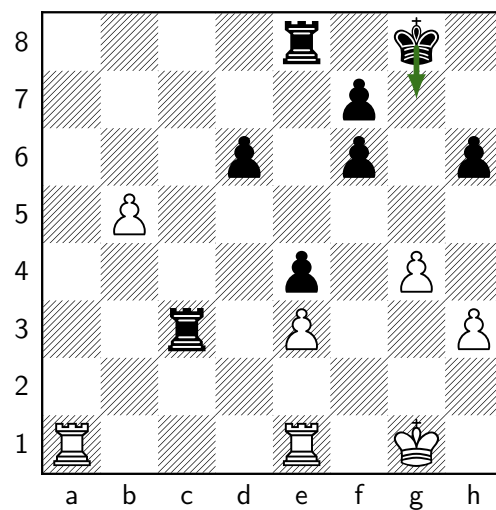


(d)

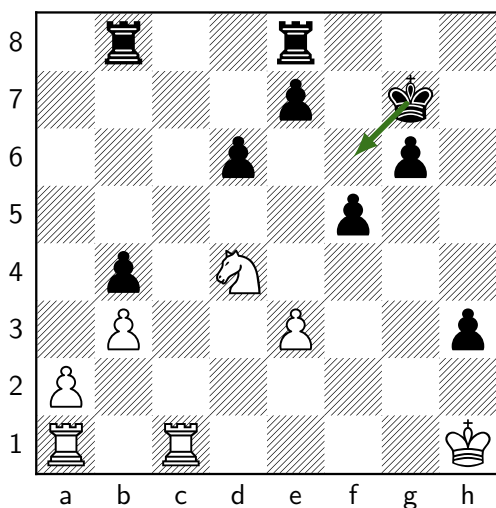
Figure 22: **Move 28, Black: f6** (f7f6). (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1907` (ply 60) [structural distance  $d=43$ , QAP obj=28]: the master played Rh8; under  $\mu$  our Pf2 plays the role of that game’s Re8 (a **cross-type** correspondence); cross-type roles: our pa4 as their Rf2, our Ra8 as their Ph2; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#102` (ply 63) [structural distance  $d=41$ , QAP obj=25]: the master played Re8; under  $\mu$  our Pf2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Bf3 as their pb3, our Ra8 as their Bh3; transferred move f7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3505` (ply 52) [structural distance  $d=41$ , QAP obj=21]: the master played Rb3; under  $\mu$  our Pf2 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Ph2, our Nb8 as their Pf3; transferred move f7f6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to f6 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



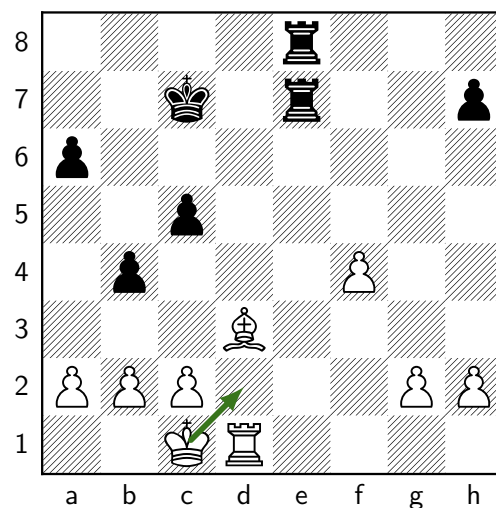
(a)



(b)



(c)



(d)

Figure 23: **Move 29, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#2185` (ply 69) [structural distance  $d=46$ , QAP obj=22]: the master played Kg7; under  $\mu$  our Kf1 plays the role of that game’s Kg1; cross-type roles: our Pg7 as their Re8, our Ra8 as their Pf7; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#946` (ply 65) [structural distance  $d=51$ , QAP obj=21]: the master played Kf6; under  $\mu$  our Kf1 plays the role of that game’s Kg2; cross-type roles: our Nb8 as their Re8, our ne3 as their Pe7; transferred move f8g8 (**endorses this move**). (d) Retrieved from `ccl1_top_p0003#1726` (ply 46) [structural distance  $d=51$ , QAP obj=20]: the master played Kd2; under  $\mu$  our Kf1 plays the role of that game’s Kc1; cross-type roles: our Nb8 as their Rd1, our Ra8 as their Pc2; transferred move f8g8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kg8 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



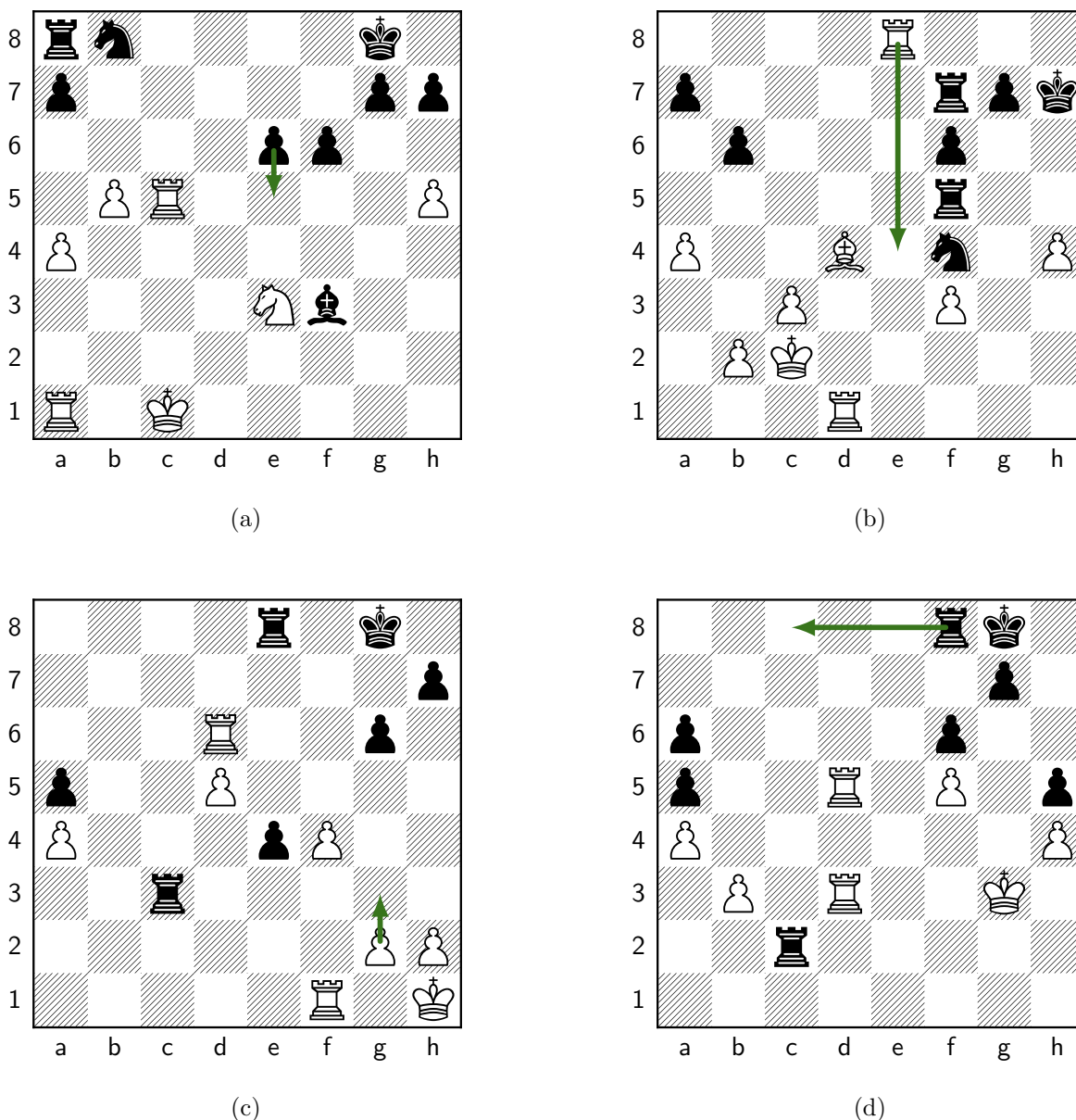
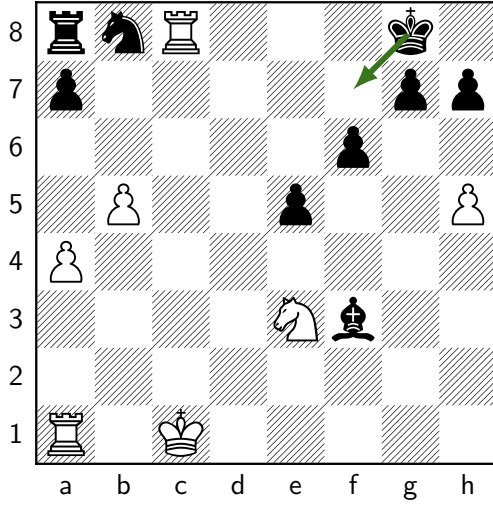
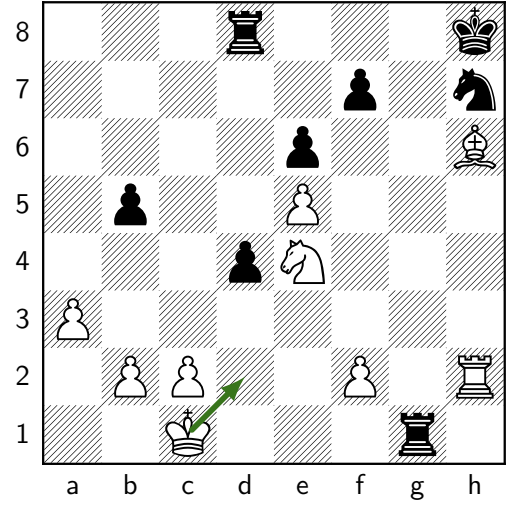


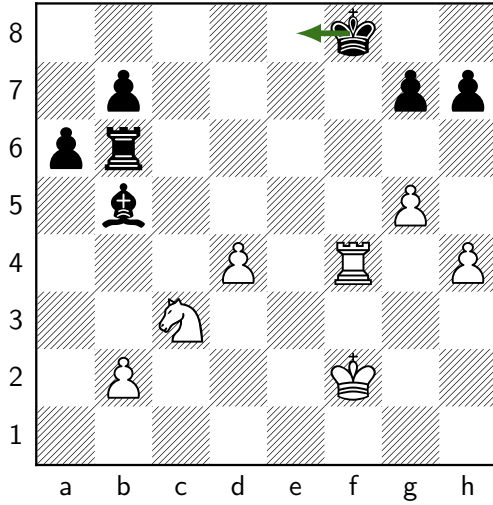
Figure 24: **Move 30, Black: e5**. (a) Our position; Melanie plays e5 (e6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2436` (ply 74) [structural distance  $d=54$ , QAP obj=35]: the master played Re4; under  $\mu$  our Pe3 plays the role of that game’s Re8 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pb2, our ra1 as their pa7; transferred move e6e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3266` (ply 64) [structural distance  $d=55$ , QAP obj=18]: the master played g3; under  $\mu$  our Pe3 plays the role of that game’s Pg2; cross-type roles: our Pa7 as their Rf1, our ne3 as their pe4; transferred move e6e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1588` (ply 71) [structural distance  $d=51$ , QAP obj=16]: the master played Rfc8; under  $\mu$  our Pe3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our rc5 as their Pa6, our Bf3 as their rd5; transferred move e6e5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to e5 (summed QAP objective 84, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



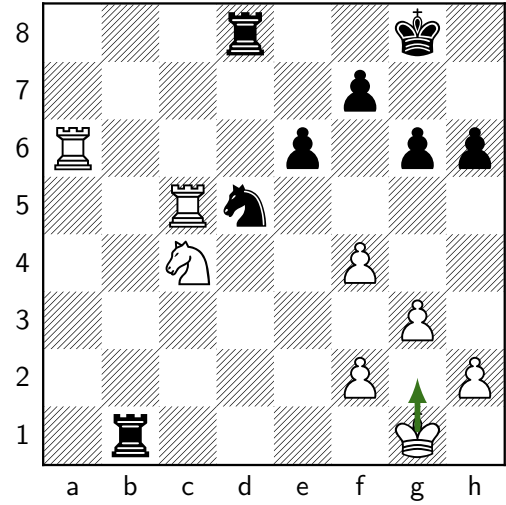
(a)



(b)

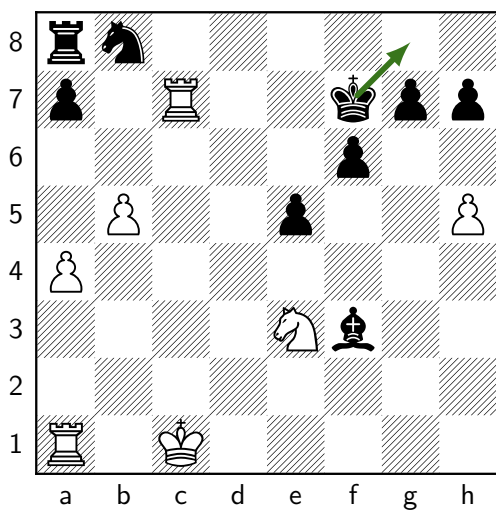


(c)

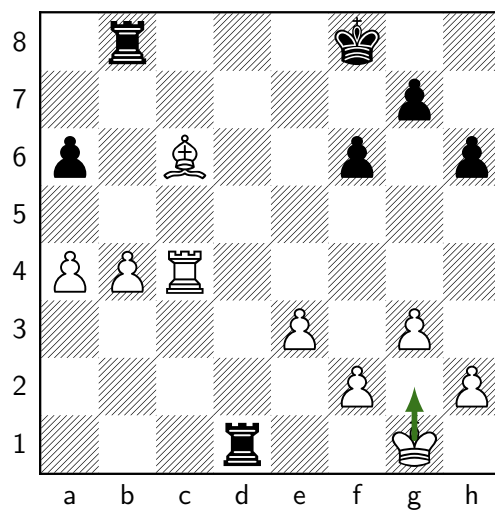


(d)

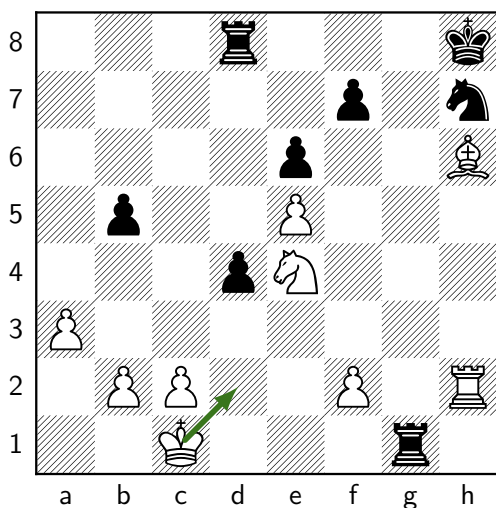
Figure 25: **Move 31, Black: Kf7**. (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2335` (ply 58) [structural distance  $d=38$ , QAP obj=73]: the master played Kd2; under  $\mu$  our Kg1 plays the role of that game's Kc1; cross-type roles: our Ra8 as their Pf2, our Nb8 as their Rh2; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1973` (ply 61) [structural distance  $d=44$ , QAP obj=69]: the master played Ke8; under  $\mu$  our Kg1 plays the role of that game's Kf1; cross-type roles: our ne3 as their Pa6, our Pe5 as their Rb6; transferred move g8f7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#349` (ply 64) [structural distance  $d=47$ , QAP obj=59]: the master played Kg2; under  $\mu$  our Kg1 plays the role of that game's Kg1; cross-type roles: our Pa7 as their Nc4, our ne3 as their Pf4; transferred move g8f7 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Kf7 (summed QAP objective 419, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



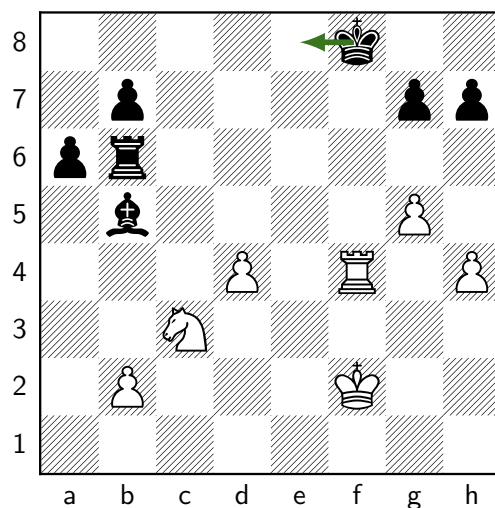
(a)



(b)

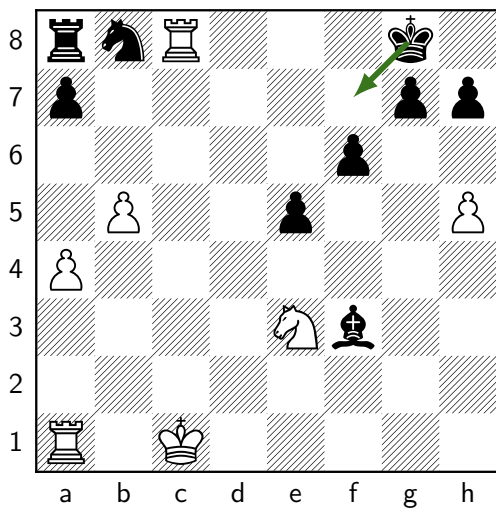


(c)

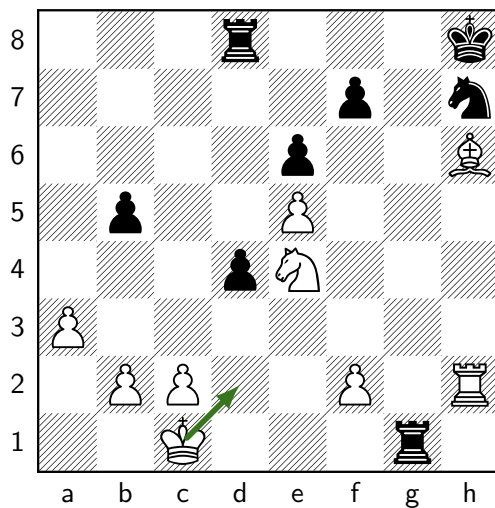


(d)

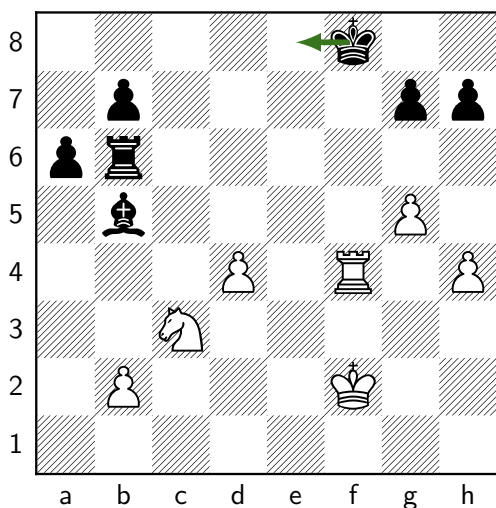
Figure 26: **Move 32, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3002` (ply 72) [structural distance  $d=41$ , QAP obj=71]: the master played Kg2; under  $\mu$  our Kf2 plays the role of that game’s Kg1; cross-type roles: our Nb8 as their Pe3, our Bf3 as their Pa4; transferred move f7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2335` (ply 58) [structural distance  $d=36$ , QAP obj=55]: the master played Kd2; under  $\mu$  our Kf2 plays the role of that game’s Kc1; cross-type roles: our Nb8 as their Pf2, our Ph7 as their Ne4; transferred move f7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1973` (ply 61) [structural distance  $d=44$ , QAP obj=53]: the master played Ke8; under  $\mu$  our Kf2 plays the role of that game’s Kf1; cross-type roles: our Nb8 as their Ph7, our Pg7 as their Rb6; transferred move f7g8 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kg8 (summed QAP objective 333, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



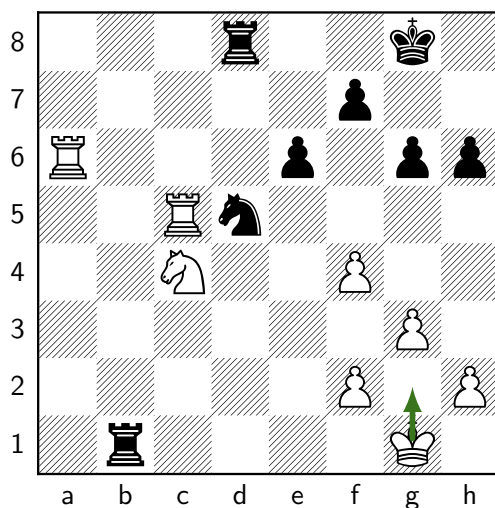
(a)



(b)

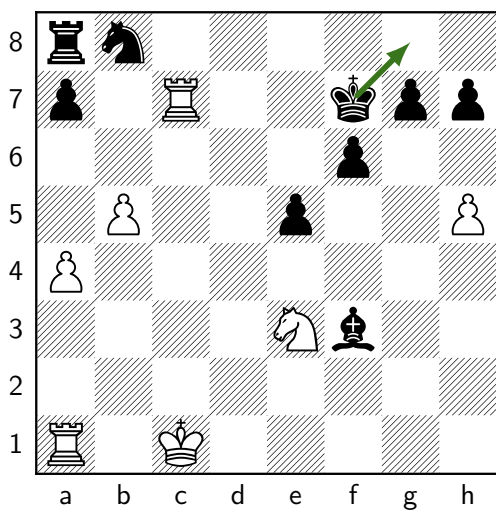


(c)

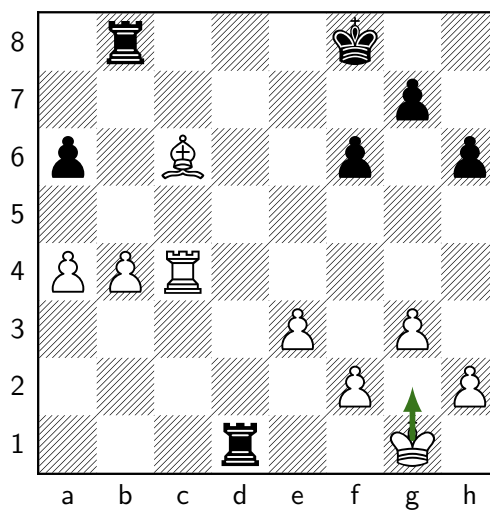


(d)

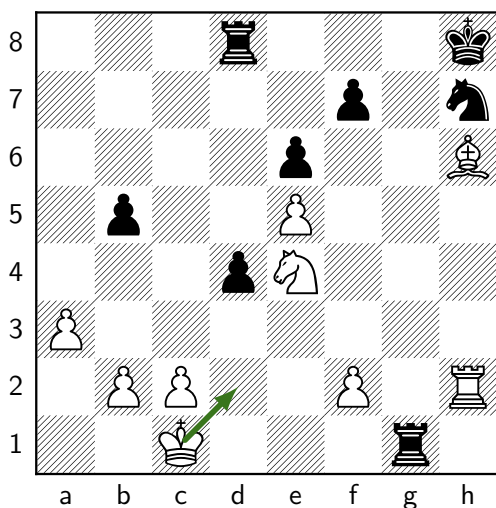
Figure 27: **Move 33, Black: Kf7**. (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2335` (ply 58) [structural distance  $d=38$ , QAP obj=73]: the master played Kd2; under  $\mu$  our Kg1 plays the role of that game’s Kc1; cross-type roles: our Ra8 as their Pf2, our Nb8 as their Rh2; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1973` (ply 61) [structural distance  $d=44$ , QAP obj=69]: the master played Ke8; under  $\mu$  our Kg1 plays the role of that game’s Kf1; cross-type roles: our ne3 as their Pa6, our Pe5 as their Rb6; transferred move g8f7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#349` (ply 64) [structural distance  $d=47$ , QAP obj=59]: the master played Kg2; under  $\mu$  our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pa7 as their Nc4, our ne3 as their Pf4; transferred move g8f7 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kf7 (summed QAP objective 432, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



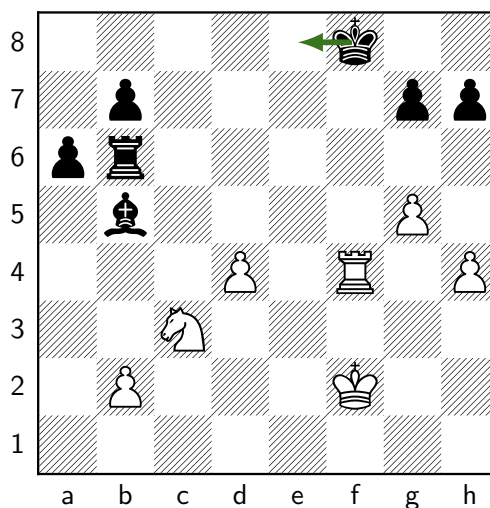
(a)



(b)



(c)



(d)

Figure 28: **Move 34, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3002` (ply 72) [structural distance  $d=41$ , QAP obj=71]: the master played Kg2; under  $\mu$  our Kf2 plays the role of that game’s Kg1; cross-type roles: our Nb8 as their Pe3, our Bf3 as their Pa4; transferred move f7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2335` (ply 58) [structural distance  $d=36$ , QAP obj=55]: the master played Kd2; under  $\mu$  our Kf2 plays the role of that game’s Kc1; cross-type roles: our Nb8 as their Pf2, our Ph7 as their Ne4; transferred move f7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1973` (ply 61) [structural distance  $d=44$ , QAP obj=53]: the master played Ke8; under  $\mu$  our Kf2 plays the role of that game’s Kf1; cross-type roles: our Nb8 as their Ph7, our Pg7 as their Rb6; transferred move f7g8 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kg8 (summed QAP objective 333, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.